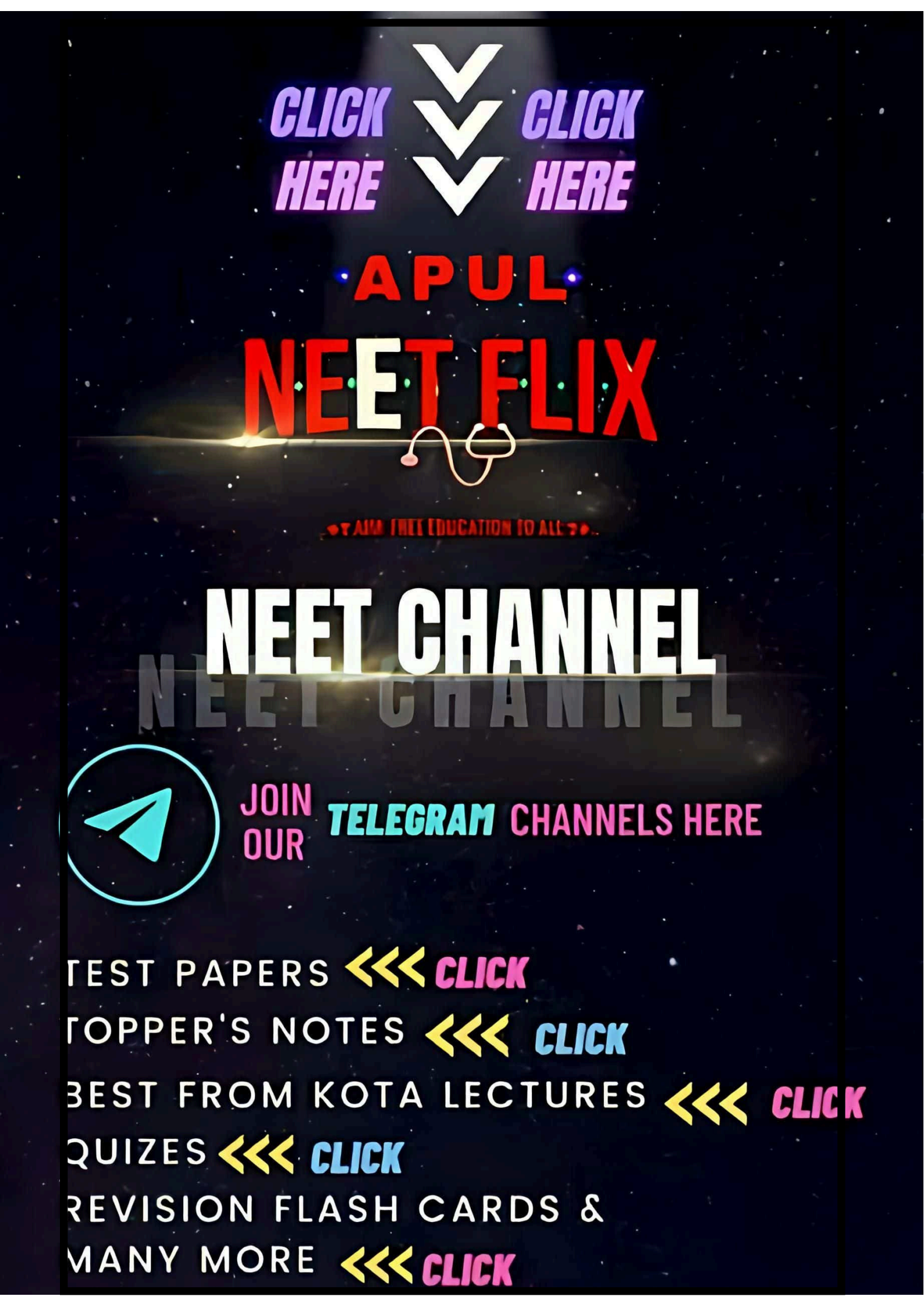

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General Organic Chemistry

Reconcile the concept :

- –I effect tends to disperse the –ve charge thus stabilized the carbanion.
- +I effect tends to intensity the –ve charge thus destabilized the carbanion.
- –I effect : $-\overset{\oplus}{O}R_2 > -\overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 >$ – NO_2 > – CN > – COOH > Halogens > – OR > – OH > – C ≡ CH > – NH_2 > – Ph > –CH = CH_2 > –H
- + I effect: $-\overset{s}{N}H > -\overset{s}{O} >$ $-C(CH_3)_3 >$ $-CH(CH_3)_2 >$ $-CH_2-CH_3 >$ – CD_3 > – CH_3 > – D > – H

1. Arrange the following in decreasing order of their stability :-

(I) $CH_3-\overset{\oplus}{C}H_2$ (II) $CH_3-\overset{\oplus}{C}H(CH_3)$

(III) $CH_3-\overset{\oplus}{C}(CH_3)(CH_3)$ (IV) $CH_2(F)-\overset{\oplus}{C}H_2$

(1) II > I > III > IV (2) IV > I > II > III
(3) IV > III > II > I (4) III > II > I > IV

CGC191

2. Which of the following represents the correct order of decreasing inductive effect:

(1) $-\overset{\oplus}{N}H_3 > -NO_2 > -Br$

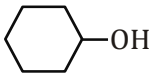
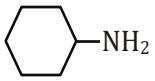
(2) $-\overset{\oplus}{N}H_3 > -Br > -NO_2$

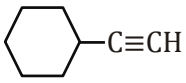
(3) $-Br > -NO_2 > -\overset{\oplus}{N}H_3$

(4) $-NO_2 > -Br > -\overset{\oplus}{N}H_3$

CGC192

3. Give the correct order of increasing acidic strength of the following compounds :

(I)  (II) 

(III) 

(1) II < I < III (2) III < II < I
(3) II < III < I (4) I < II < III

CGC193

4. Most stable anion is :

(1) $CH_3-CH_2-\overset{\ominus}{N}H$ (2) $CH_3-\overset{CH_3}{\overset{|}{C}}(CH_3)-\overset{\ominus}{O}$

(3) $CH_3-\overset{\ominus}{C}H_2$ (4) $CH_2=\overset{\ominus}{C}H$

CGC194

5. Arrange the following in correct order of their stability ?

(I) $CH\equiv\overset{\ominus}{C}$ (II) $CH_2=\overset{\ominus}{C}H$ (III) $CH_3-\overset{\ominus}{C}H_2$

(1) I > II > III (2) III > II > I
(3) II > III > I (4) II > I > III

CGC195

6. Consider the given compounds :

(a) $CH_3 - CH_2 - NH_2$ (b) $CH_3 - CH = NH$
(c) $CH_3 - C \equiv N$ (d) $C_2H_5 - NH - C_2H_5$

Arrange basicity of these compounds in decreasing order :

(1) d > a > b > c (2) a > b > c > d
(3) a > d > b > c (4) d > a > c > b

CGC196

7. Arrange the following acid in decreasing order of their acidic strength :

(I) CH_3-OH (II) CH_3CH_2-OH

(III) $CH_3-CH(CH_3)-OH$ (IV) $CH_3-\overset{CH_3}{\overset{|}{C}}(CH_3)-OH$

(1) II > III > IV > I (2) I > II > III > IV
(3) I > III > II > IV (4) II > IV > III > I

CGC197

8. Arrange the compounds in order of decreasing acidic strength :

(I) $ClCH_2-CH_2-OH$
(II) Cl_2CH-CH_2-OH
(III) $Cl-CH_2-CH_2-CH_2-OH$
(IV) $CH_3-CH_2-CH_2-OH$

(1) I > II > III > IV (2) IV > III > II > I
(3) II > I > IV > III (4) II > I > III > IV

CGC198

9. Arrange the following compound in basic strength :

(A) $CH_3-\ddot{N}H_2$ (B) $(CH_3)_2NH$ (C) $(CH_3)_3N$

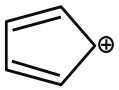
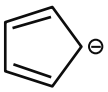
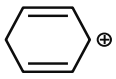
(1) A > B > C (2) B > A > C
(3) A > C > B (4) C > A > B

CGC199

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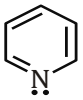
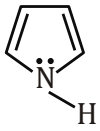
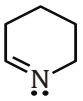
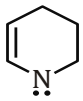
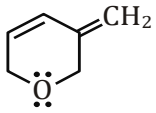
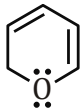
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10. Which of the following is aromatic :-

(1) 
(2) 
(3) 
(4) 

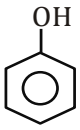
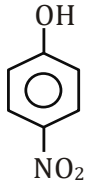
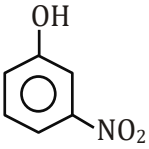
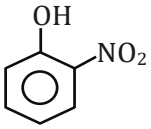
CGC200

11. In which of the following cases lone-pair is involved in resonance :

(1) 
(2) 
(3) 
(4) 
(5) $CH_2=CH-CH=\ddot{N}H$
(6) $CH_2=CH-\ddot{\underset{\cdot\cdot}{O}}-CH=CH_2$
(7) $CH_2=CH-C\equiv N:$
(8) 
(9) 

CGC201

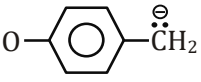
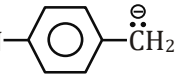
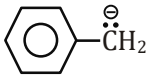
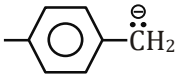
12. Arrange the following in decreasing order of acidic strength:

(I) 
(II) 
(III) 
(IV) 

(1) I > II > III > IV (2) IV > III > II > I
(3) II > IV > III > I (4) IV > II > III > I

CGC202

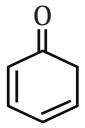
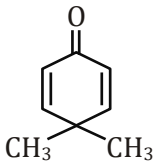
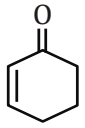
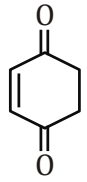
13. Consider the following carbanions :

(I) 
(II) 
(III) 
(IV) 

Correct decreasing order of stability is :
(1) II > III > IV > I (2) III > IV > I > II
(3) IV > I > II > III (4) I > II > III > IV

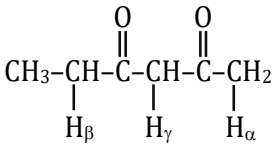
CGC203

14. Which cannot show tautomerism ?

(1) 
(2) 
(3) 
(4) 

CGC204

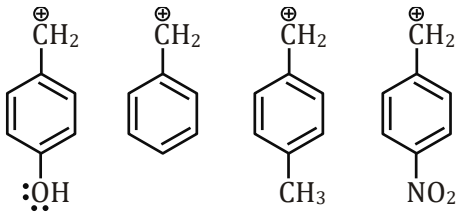
15. The order of acidic strength of the hydrogen atom (H_α, H_β, H_γ) in the given molecule is :



(1) $H_\gamma > H_\alpha > H_\beta$ (2) $H_\gamma > H_\beta > H_\alpha$
(3) $H_\alpha > H_\beta > H_\gamma$ (4) $H_\beta > H_\alpha > H_\gamma$

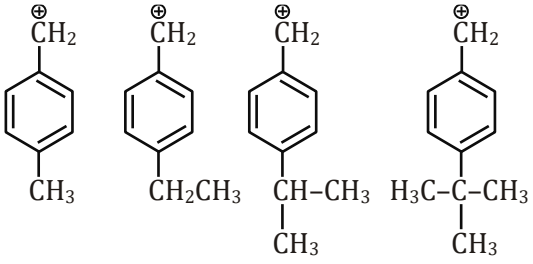
Arrange the following in decreasing order as indicated.

CGC205

16. 
(I) (II) (III) (IV)

[Stability order]

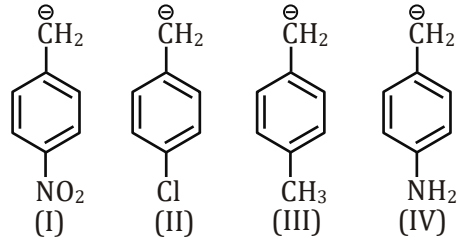
CGC206

17. 
(I) (II) (III) (IV)

[Stability order]

CGC207

18.

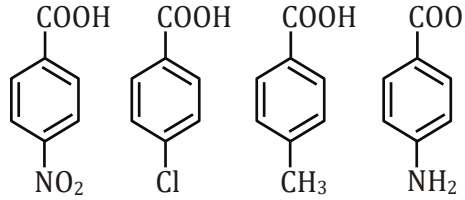


(I) (II) (III) (IV)

[Stability order]

CGC208

19.

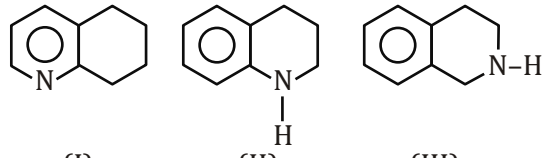


(I) (II) (III) (IV)

[Acidic strength]

CGC209

20.



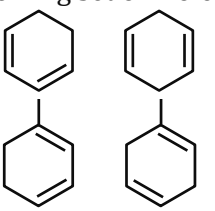
(I) (II) (III)

[Basic strength]

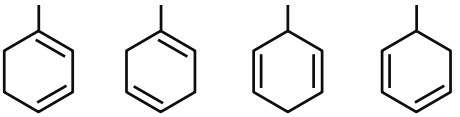
CGC210

21. What is order of heat of hydrogenation for the following set of molecules :-

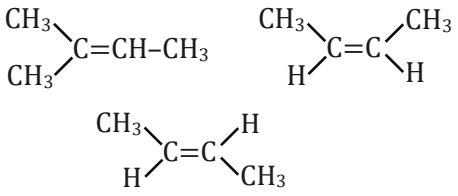
(1)



(2)



(3)



CGC211

22. The order of the –I effect orbitals in :

(1) $sp^3 > sp^2 > sp$ (2) $sp^2 > sp^3 > sp$

(3) $sp > sp^2 > sp^3$ (4) $sp^3 > sp > sp^2$

CGC212

23. The number of electron present in the valence shell of carbon of carbanion bearing –ve charge

(1) 8 (2) 7 (3) 6 (4) 4

CGC213

24. Arrange the following in the stability order :

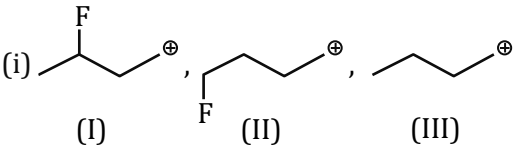
(1) $CH_3–C\equiv\overset{\ominus}{C}$ (2) $CH_3–CH=\overset{\ominus}{C}H$

(3) $CH_3–\overset{\ominus}{C}H–CH_3$ (4) $CH_3–CH_2–\overset{\ominus}{C}H_2$

CGC214

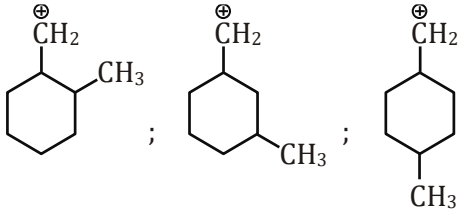
25. Arrange given carbocations in decreasing order of their stability:

(i)



(I) (II) (III)

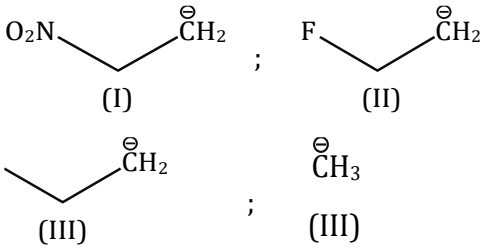
(ii)



(I) (II) (III)

CGC215

26. Arrange given anions in decreasing order of their stability



(I) (II) (III) (III)

CGC216

27. Arrange given compounds in decreasing order of their K_a value:

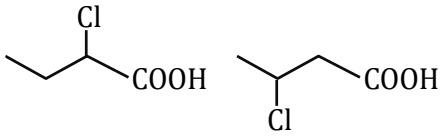
(i) HCOOH ; CH_3COOH ;

(I) (II)

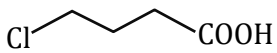
C_2H_5COOH ; C_3H_7 –COOH

(III) (IV)

(ii)

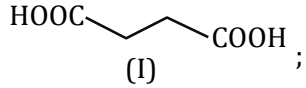


(I) (II)

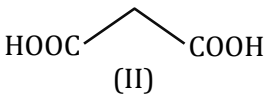


(III)

(iii)



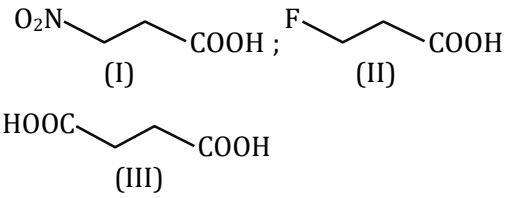
(I) ;



(II)

HOOC–COOH

(III)

(iv) 

CGC217

28. Which among the given acid has lowest pKa value

(1) Chloroacetic acid

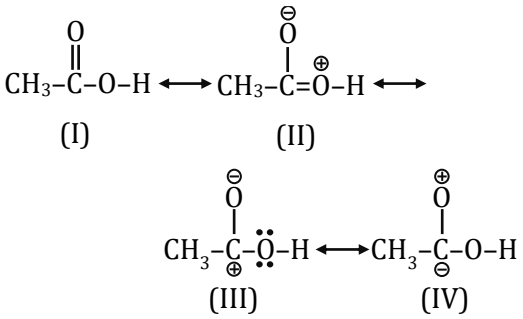
(2) Bromoacetic acid

(3) Nitroacetic acid

(4) Cyanoacetic acid

CGC218

29. Acetic acid is considered as resonance hybrid



Which of the following order is correct for the stability of four contributing structure :

(1) I > II > III > IV

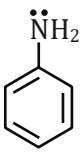
(2) I > II > IV > III

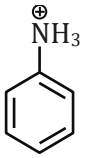
(3) I > III > II > IV

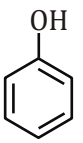
(4) I > IV > III > II

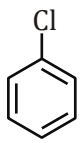
CGC219

30. In which of the following molecule group present on ring is not involved in resonance :

(1) 

(2) 

(3) 

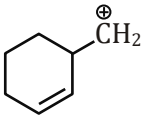
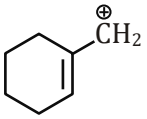
(4) 

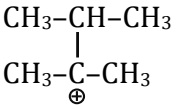
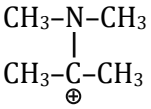
CGC220

31. In each of the following pairs of ions which ion is more stable :

(i) (I) $C_6H_5-\overset{\oplus}{C}H_2$ and (II) $CH_2=CH-\overset{\oplus}{C}H_2$

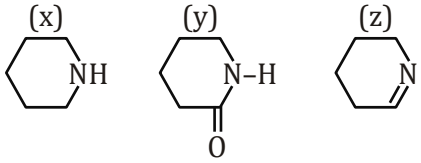
(ii) (I) $CH_3-\overset{\oplus}{C}H_2$ and (II) $CH_2=\overset{\oplus}{C}H$

(iii) (I)  and (II) 

(iv) (I)  and (II) 

CGC221

32. The correct order of decreasing basic strengths of x, y and z is :

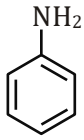


(1) x > y > z (2) x > z > y

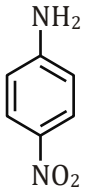
(3) y > x > z (4) y > z > x

CGC222

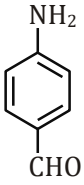
33. Consider the following compounds :

(1) 

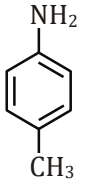
;

(2) 

;

(3) 

;

(4) 

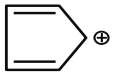
Arrange these compounds in decreasing order of their basicity :

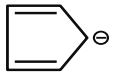
(1) 1 > 2 > 3 > 4 (2) 2 > 3 > 1 > 4

(3) 4 > 1 > 3 > 2 (4) 4 > 1 > 2 > 3

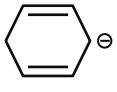
CGC223

34. Which shows aromatic character :

(1) 

(2) 

(3) 

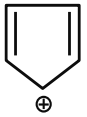
(4) 

CGC224

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35. Among the following the aromatic compound is

(1) 

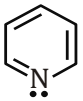
(2) 

(3) 

(4) 

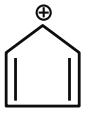
CGC225

36. Which of the following is anti-aromatic species :

(1) 

(2) 

(3) 

(4) 

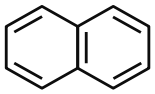
CGC226

37. Which of the following is non-aromatic :

(1) 

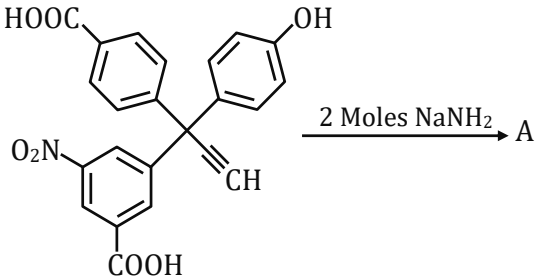
(2) 

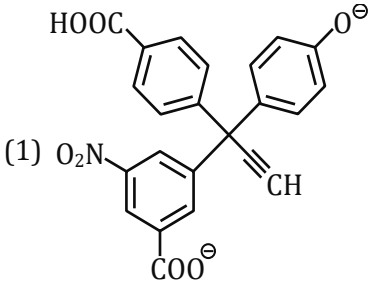
(3) 

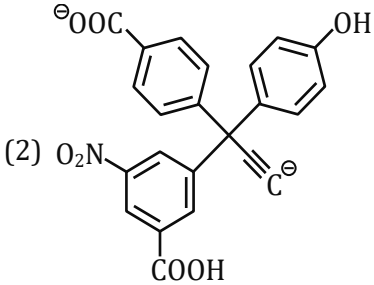
(4) 

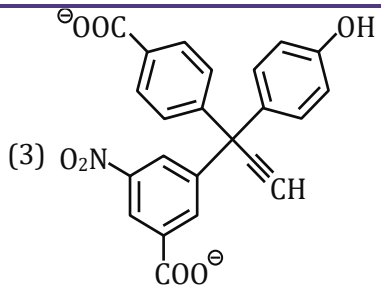
CGC227

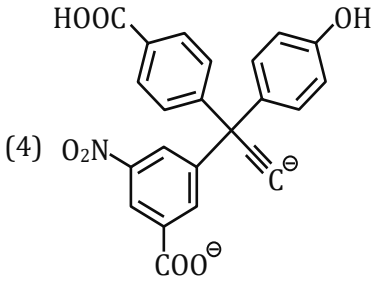
38. Product (A) will be :-



(1) 

(2) 

(3) 

(4) 

CGC228

39. Which of the following represents the correct order of stability of the given carbocations ?



(1) III > I > II (2) I > III > II

(3) III > II > I (4) II > III > I

CGC229

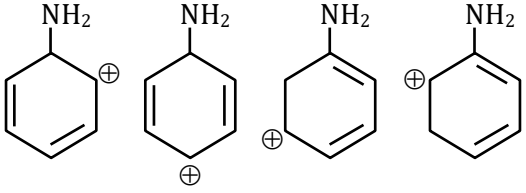
40. Which of the following group has maximum hyperconjugation effect but minimum inductive effect

(1) $-CH_3$ (2) $-CH_2-CH_3$

(3) $-CD_3$ (4) $-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$

CGC230

41. The most stable carbocation for the following is:



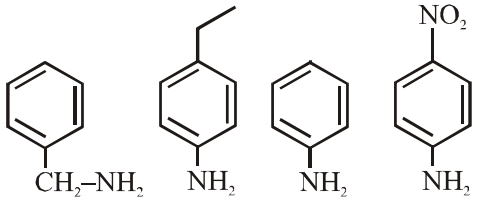
(a) (b) (c) (d)

(1) c (2) d (3) b (4) a

CGC231

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42. Correct Decreasing order of pK_b is:

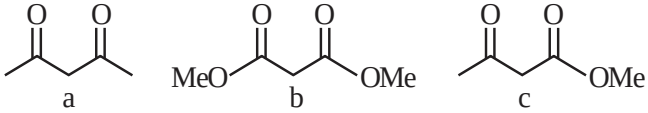


(1) (2) (3) (4)

(1) 1 > 2 > 3 > 4
(2) 4 > 3> 2 > 1
(3) 2 > 1 > 3 > 4
(4) 1 > 2 > 4 > 3

CGC232

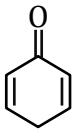
43. Which will undergo deprotonation most readily in basic medium?

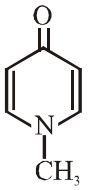


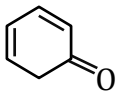
(1) a only (2) b only
(3) c only (4) None of above

CGC233

44. Which among these can exhibit tautomerism?

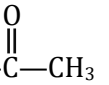
(a) 

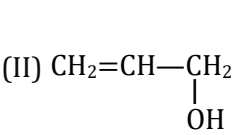
(b) 

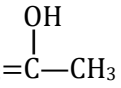
(c) 

(1) a only (2) b only
(3) c only (4) a and c

CGC234

45. (I) 

(II) 

(III) 

Which among these are tautomers?
(1) I and II (2) I and III
(3) II and III (4) All of these

CGC235

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**4**	**1**	**3**	**2**	**1**	**1**	**2**	**4**	**2**	**2**
Question	11	12	13	14	15	16			17	
Answer	**2, 4, 6, 9**	**3**	**1**	**2**	**1**	**I > III > II > IV**			**I > II > III > IV**	
Question	18		19		20	21				
Answer	**I > II > III > IV**		**I > II > III > IV**		**III > I > II**	**(1) II > I (2) III > II > IV > I (3) II > III > I**				
Question	22	23	24		25	26				
Answer	**3**	**1**	**I > II > IV > III**		**(i) III > II > I (ii) I > II > III**	**(i) I > II > IV > III**				
Question	27					28	29		30	
Answer	**(i) I > II > III > IV (ii) I > II > III (iii) III > II > I (iv) I > III > II**					**3**	**1**		**2**	
Question	31				32	33	34	35	36	37	38
Answer	**(i) I > II (ii) I > II (iii) II > I (iv) II > I**				**2**	**3**	**2**	**1**	**4**	**2**	**3**
Question	39	40	41	42	43	44	45				
Answer	**1**	**1**	**1**	**2**	**1**	**4**	**2**				

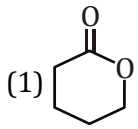
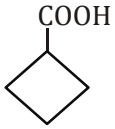
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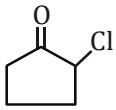
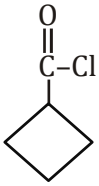
Isomerism

1. $CH_3CH_2CH(CHO)CH_2CH_3$ and $CH_3CH(CHO)\,CH_2CH_2CH_3$ are :-
(1) Chain isomer (2) Position isomer
(3) Metamerism (4) Functional isomer

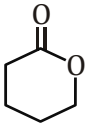
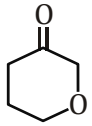
CIS077

2. Find out relation between the following compounds:

(1)  and 

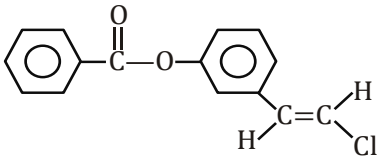
(2)  and 

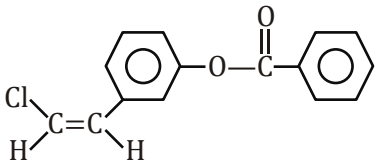
(3) $CH_3CH_2CH_2\underset{\underset{COOH}{|}}{C}HCH_3$ and $CH_3CH_2\underset{\underset{COOH}{|}}{C}HCH_2CH_3$

(4)  and 

CIS078

3. Isomerism shown by the following compounds

 and

 is :-

(1) Functional group isomerism
(2) Geometrical isomerism
(3) Metamerism
(4) Position isomerism

CIS079

4. How many structural isomers are possible when one H-atom of anthracene is replaced by chlorine:-
(1) 3 (2) 7 (3) 4 (4) 6

CIS080

5. How many amines are possible with the molecular formula $C_4H_{11}N$ (structural isomers only) :-
(1) 6 (2) 7 (3) 8 (4) 9

CIS081

6. How many total structural isomers of 3° amines are possible with the molecular formula $C_6H_{15}N$
(1) 4 (2) 7 (3) 6 (4) 5

CIS082

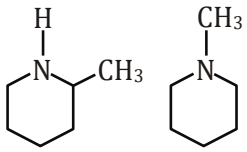
7. Only two isomeric monochloro derivatives are possible for (excluding stereo) :-
(1) n-Butane
(2) 2,2-Dimethylpentane
(3) Benzene
(4) Neopentane

CIS083

8. How many benzenoid isomers are possible for molecular formula C_7H_8O :-
(1) 3 (2) 4 (3) 5 (4) 6

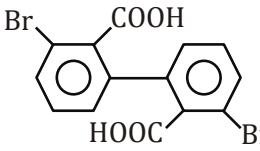
CIS084

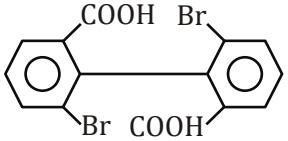
9. The relation between given compound is :-



(1) Chain isomerism
(2) Position isomerism
(3) Functional isomerism
(4) Metamerism

CIS085

10.  and

 are :-

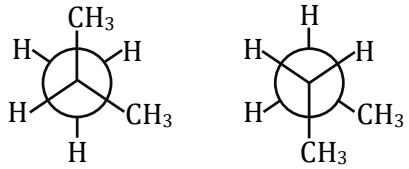
(1) Chain isomers
(2) Geometrical isomers
(3) Functional isomers
(4) Position isomers

CIS086

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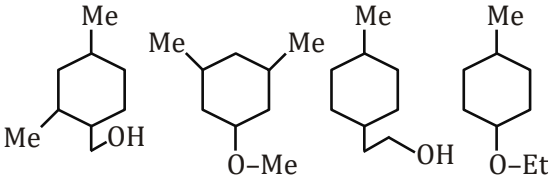
11. The structure below are :-



(1) Not isomers
(2) Conformational isomers
(3) Cis-trans isomers
(4) Structural isomers

CIS087

12. Which of the following is not the correct relationship :



(I) (II) (III) (IV)

(1) I and IV are positional isomers
(2) I and II are functional isomers
(3) I and III are chain isomers
(4) II & IV are metamers

CIS088

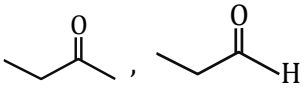
13. Number of structural isomers possible for dibromo derivative of an alkene with molar mass 186 gm mol^{-1} is (atomic mass of Br = 80)
(1) 2 (2) 3 (3) 4 (4) 6

CIS089

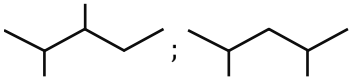
14. How many alkenes (only structural isomers) are possible with the molecular formula C_4H_8
(1) 4 (2) 3 (3) 5 (4) 6

CIS090

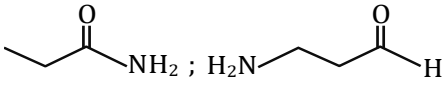
15. Which of the following is correct :-

(1) 

Functional group isomers

(2) 

Position isomers

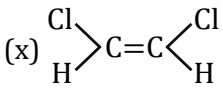
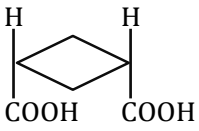
(3) 

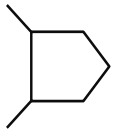
Position and functional group isomers

(4) All of these

CIS091

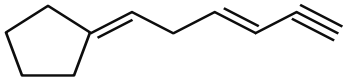
16. Which of the following compounds can show geometrical isomerism :
(i) Pent-2-ene (ii) Pent-3-en-1-yne
(iii) Acetaldoxime (iv) But-2-yne
(v) 2-chlorobutane (vi) Butenyne
(vii) 1-chloropropene
(viii) 2-Methylbut-2-ene
(ix) $CH_3–CH=CH–CH_3$

(x)  (xi) 

(xii) 

CIS092

17. How many geometrical isomers are possible for the given compound



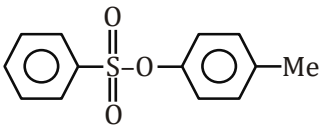
(1) 2 (2) 4 (3) 6 (4) 8

CIS093

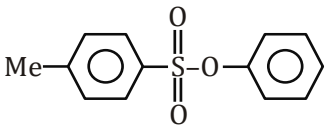
18. Minimum C-atom required for aliphatic 2° amine to show chain isomerism are :-
(1) 4 (2) 6 (3) 3 (4) 5

CIS094

19. Isomerism shown by



and



(1) Chain isomerism
(2) Positional isomerism
(3) Metamerism
(4) Functional group isomerism

CIS095

20. Branched chain isomer of a hydrocarbon having molecular mass 72u gives only one isomeric alkyl halide after monosubstitution. The hydrocarbon is :-
(1) neohexane (2) t-butyl chloride
(3) neopentane (4) isooctane

CIS096

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21. How many structural isomers are possible of $C_5H_{12}O$:-

(1) 5 (2) 8 (3) 13 (4) 14

CIS097

22. What is relation between following compounds

$CH_3-CH_2-\underset{\underset{CN}{|}}{CH}-CH_2-CHO$

and

$CH_3-\underset{\underset{CN}{|}}{CH}-CH_2-CH_2-CHO$

(1) positional isomers (2) chain isomers

(3) optical isomers (4) metamers

CIS098

23. No. of structural isomeric alkenes (molecular formula=C_6H_{12}) which all give n-hexane on hydrogenation in presence of metal catalyst

(1) 2 (2) 3 (3) 4 (4) 5

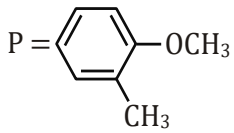
CIS099

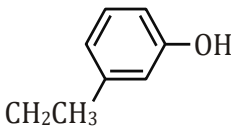
24. How many isomeric geminal dihalides are possible for the compound having M.F. $C_3H_6Cl_2$

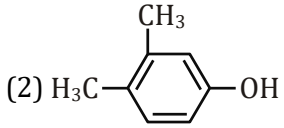
(1) 1 (2) 2 (3) 3 (4) 4

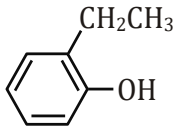
CIS100

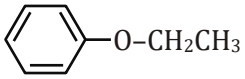
25. Which is the metamer of the compound P ?

P = 

(1) 

(2) 

(3) 

(4) 

CIS101

26. Which of the following compound have minimum dipole moment :

(1) $CH_3-CH_2-C\equiv CH$ (2) $CH_3-C\equiv C-CH_3$

(3) $CH_3-CH=CH_2$ (4) $CH_3-C\equiv CH$

CIS102

27. In which compound cis-trans nomenclature cannot be used :

(1) $CH_3-CH=CH-COOH$

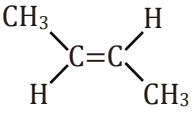
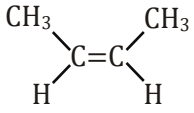
(2) $C_6H_5-CH=CH-C_6H_5$

(3) $C_6H_5-CH=CHD$

(4) $C_6H_5-CH=C(CH_3)(Cl)$

CIS103

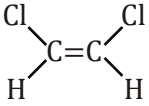
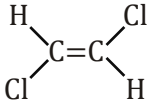
28. Which is true for the following isomeric forms I & II respectively :

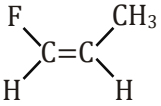
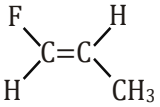
(I)  (II) 

	Dipole moment	Boiling point	Melting point	Stability
(1)	I > II	I > II	II > I	I > II
(2)	II > I	II > I	II > I	II > I
(3)	I > II	I > II	I > II	I > II
(4)	II > I	II > I	I > II	I > II

CIS104

29. Compare dipole moment of the following

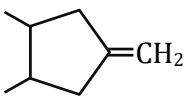
(A)  ; 

(B)  ; 

(C) $CH_3-CH_2-C\equiv CH$; $CH_3-C\equiv C-CH_3$

CIS105

30. GI is shown by :

(1)  (2) $CH_3CH=C=CHCH_3$

(3)  (4) $CH_3CH=C=C=CH_2$

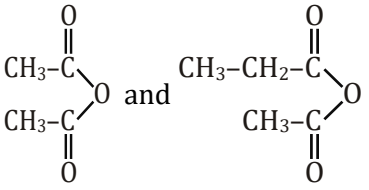
CIS106

31. Which of the following is a pair of metamers

(1) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH$ and $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_3$

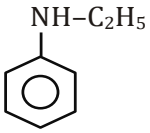
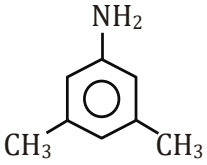
(2) $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_3$ and $H-\overset{O}{\overset{\|}{C}}-O-CH_2CH_3$

(3) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$ and $H-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_3$

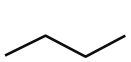
(4)  and

CIS107

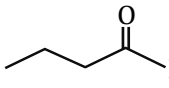
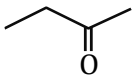
32. Identify relation that is incorrect

(1)  and 

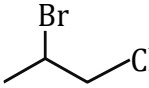
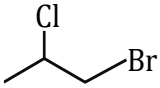
functional isomers

(2)  and 

Chain isomers

(3)  and 

Metamers

(4)  and 

Position isomers

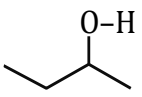
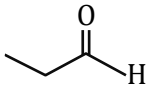
CIS108

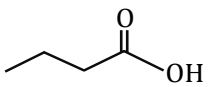
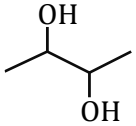
33. How many structural isomeric alkenes are possible with molecular formula C_5H_{10} that does not show geometrical isomerism

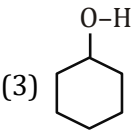
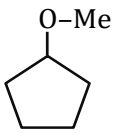
(1) 2 (2) 4 (3) 5 (4) 3

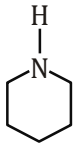
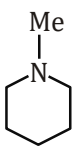
CIS109

34. Which of the following is pair of functional isomers ?

(1)  , 

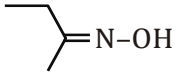
(2)  , 

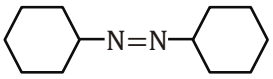
(3)  , 

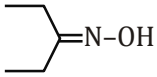
(4)  , 

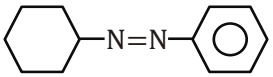
CIS110

35. Which of the following does not show geometrical isomerism

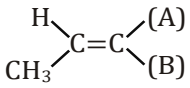
(1) 

(2) 

(3) 

(4) 

CIS111

36. For compound,  which statement is correct

(1) If (A) and (B) have same molecular formula then compound never show geometrical isomerism.

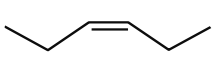
(2) If (A) and (B) have same molecular formula then compound always show geometrical isomerism.

(3) If (A) and (B) have same molecular formula then compound may or may not show geometrical isomerism.

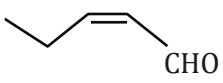
(4) None of the above

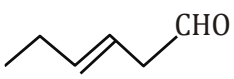
CIS112

37. Identify structure of trans-2-hexenal.

(1) 

(2) 

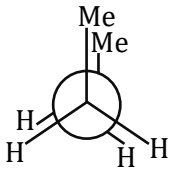
(3) 

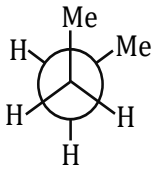
(4) 

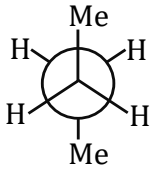
CIS113

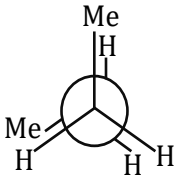
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38. Which is the highest energy conformation of butane :-

(1) 

(2) 

(3) 

(4) 

CIS114

39. Which conformer of the given compound is most stable (rotation about C–C bond) :-

$HO–CH_2–CH_2–NO_2$

(1) Staggered

(2) Gauche

(3) Fully eclipsed

(4) Partially eclipsed

CIS115

40. **Statement-I :** Dihedral angle in Gauche form of n-Butane is 60°

Statement-II : Gauche form of n-Butane is more stable than staggard form

(1) Both statement I and statement II are incorrect.

(2) Both statement-I and statement-II are correct.

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct.

CIS116

41. Which of the following compound show metamerism ?

(1) $CH_3—CO—C_2H_5$

(2) $CH_3—O—CH_3$

(3) $C_2H_5—S—C_2H_5$

(4) $CH_3—O—C_2H_5$

CIS117

ANSWER KEY

Question	1	3	4	5	6	7	8	9	10	11	12	13	14	15	17
Answer	1	2	1	3	2	1	3	3	4	4	1	1	2	2	1
Question	18	19	20	21	22	23	24	25	26	27	28	30	31	32	33
Answer	1	3	3	4	2	2	2	4	2	4	4	1	2	3	2
Question	34	35	36	37	38	39	40	41							
Answer	3	3	3	2	1	2	3	3							

2. (i) Functional Group Isomerism
(ii) Functional Group Isomerism
(iii) Chain Isomerism
(iv) Functional Isomerism

16. (i), (ii), (iii), (vii), (ix), (x), (xi), (xii)

29. (A) i > ii (B) ii > i (C) i > ii

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Optical Isomerism

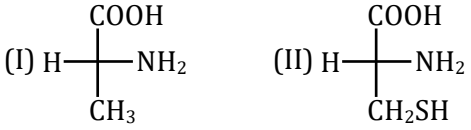
1. Match Column - I with Column - II and select the correct answer using the codes given below:

Column-I		Column-II	
(a)	Meso compound	(p)	An equimolar mixture of enantiomers
(b)	Enantiomers	(q)	Stereoisomers that are not mirror images
(c)	Diastereomers	(r)	Non superimposable mirror image
(d)	Racemates (Racemic mixture)	(s)	An optically inactive compound whose molecule is achiral even though they contain chiral centres

(1) A-S, B-R, C-Q, D-P
(2) A-S, B-R, C-P, D-Q
(3) A-R, B-S, C-Q, D-P
(4) A-R, B-S, C-P, D-Q

COI039

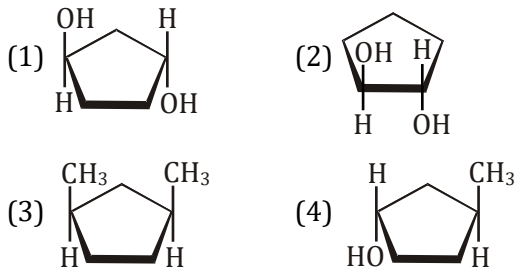
2. The correct absolute configuration assigned for compound (I) and (II) respectively is :



(1) R, R (2) R, S
(3) S, S (4) S, R

COI040

3. Which of the following cyclopentane derivative is optically inactive ?



COI041

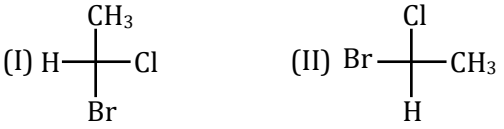
4. Match Column - I with Column - II and select the correct answer using the codes given below :

Column-I		Column-II	
(a)	Constitutional	(p)	Stereoisomers that are not mirror image isomer
(b)	Stereoisomers	(q)	Isomers that have same constitution but differ in the arrangement of their atoms in space
(c)	Enantiomers	(r)	Isomers that differ in the order in which their atoms are connected
(d)	Diastereoisomers	(s)	Stereoisomers that are related as an object and its non superimposable mirror image

(1) A-P, B-Q, C-S, D-R
(2) A-P, B-Q, C-R, D-S
(3) A-R, B-Q, C-S, D-P
(4) A-R, B-Q, C-P, D-S

COI042

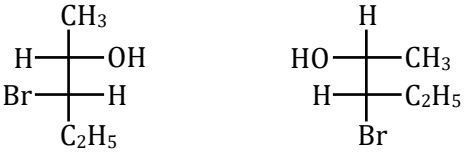
5. The following structures respresent a pair of:



(1) enantiomers (2) diastereomers
(3) meso compounds (4) homomers

COI043

6.



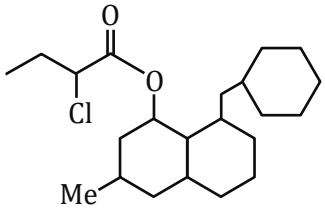
The molecules represented by the above two structures are :
(1) identical (2) enantiomers
(3) diastereomers (4) epimers

COI044

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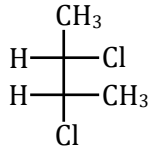
7. How many chiral centres are present in the following compound :-

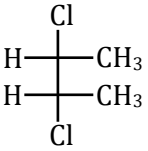


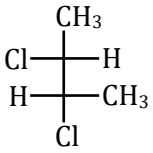
(1) 4 (2) 5 (3) 6 (4) 7

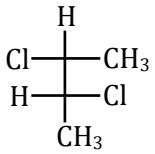
COI045

8. Which of the following structure does not represent meso isomer :-

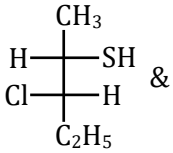
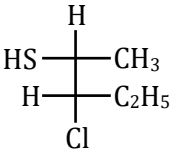
(1) 

(2) 

(3) 

(4) 

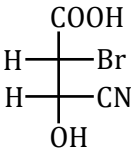
COI046

9.  &  are :-

(1) Identical (2) Enantiomers
(3) Diastereomers (4) Positional isomers

COI047

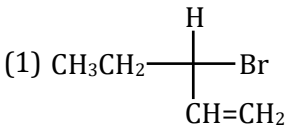
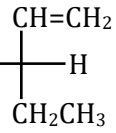
10. Configurations of the two centres in the following molecule are :

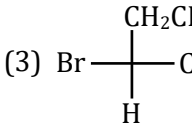
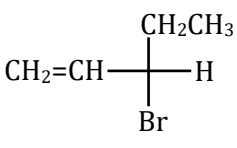


(1) (2R, 3S) (2) (2R, 3R)
(3) (2S, 3S) (4) (2S, 3R)

COI048

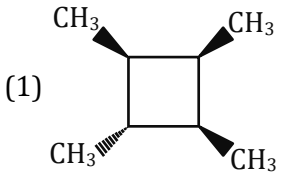
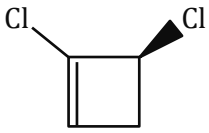
11. Which has R-configuration

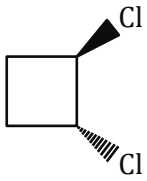
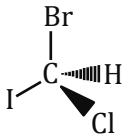
(1)  (2) 

(3)  (4) 

COI049

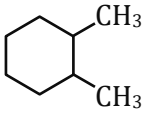
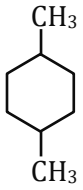
12. In which of the following compound have plane of symmetry ?

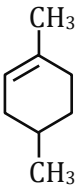
(1)  (2) 

(3)  (4) 

COI050

13. Which of the following compound will not show optical isomerism

(1)  (2) 

(3)  (4) All of these

COI051

14. d-Tartaric acid and Meso tartaric acid are-
(1) Enantiomers
(2) Diastereomers
(3) Not isomers
(4) Geometrical isomers

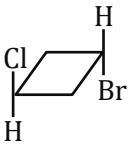
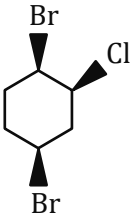
COI052

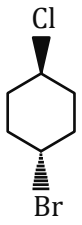
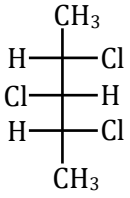
15. Number of chiral-C present in CH_3-$(CHOH)_5$-CH_3 ?

(1) 4 (2) 5 (3) 3 (4) 6

COI053

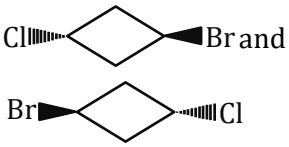
16. Which have no POS ?

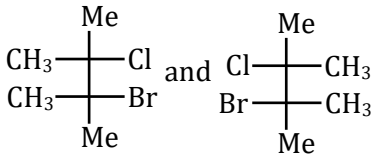
(1)  (2) 

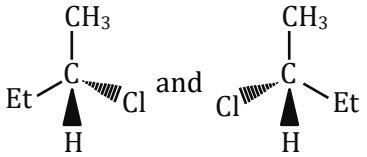
(3)  (4) 

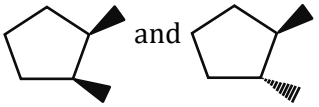
COI054

17. Which are enantiomers ?

(1)  and

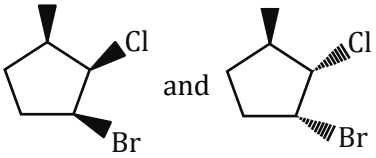
(2) 

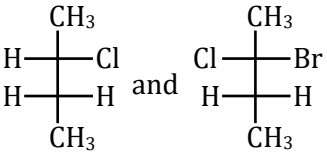
(3) 

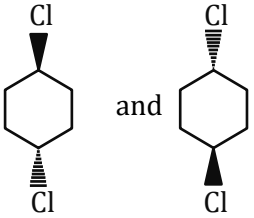
(4) 

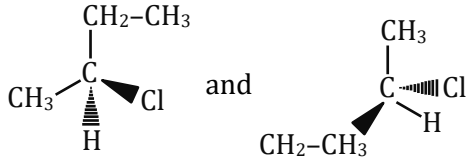
COI055

18. Which are diastereomers ?

(1) 

(2) 

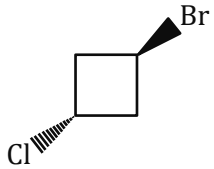
(3) 

(4) 

COI056

19. Which compound can exist in Meso ?

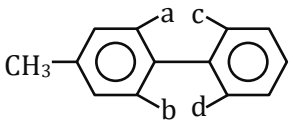
(1) Pentane-2, 3-diol

(2) 

(3) 2-Bromo-3-Chlorobutane

(4) 1,2-Dimethyl cyclopentane

COI057

20. 

shows optical isomerism if

(1) a = $-CH_3$, b = $-CH_3$, C = $-NO_2$, d = $-COOH$

(2) a = Cl, b = Br, c = $-NO_2$, d = $-CN$

(3) a = $-CH_3$, b = $-CH_2-CH_3$, c = Br, d = I

(4) a = H, b = Br, c = $-CH_3$, d = $-NO_2$

COI058

21. Which is not coplanar and shows optical isomerism ?

(1) $CH_2 = CH - CH = CH_2$

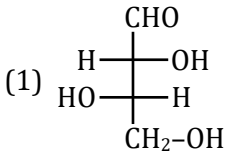
(2) $CH_2 = C = CH_2$

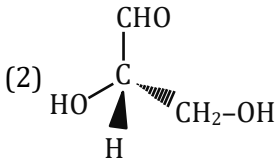
(3) $CH_3 - CH = C = CH - CH_3$

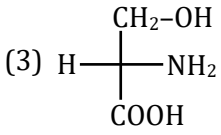
(4) $CH_2 = C = C = CH_2$

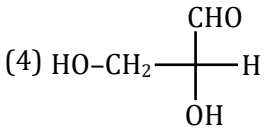
COI059

22. Which has D-configuration?

(1) 

(2) 

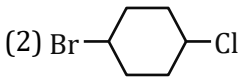
(3) 

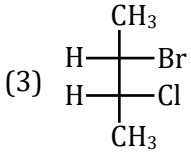
(4) 

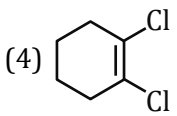
COI060

23. Which compound can show enantiomerism ?

(1) 2,3-Dimethyl butane

(2) 

(3) 

(4) 

COI061

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24. Incorrect statement about Enantiomers ?

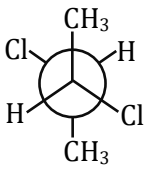
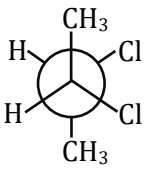
(1) They are optically active

(2) Equal mole mixture of enantiomers is called as racemic mixture.

(3) One is d and another is ℓ

(4) D & L-Glyceraldehyde are enantiomers

COI062

25.  and  are ?

(1) Enantiomers (2) Conformations

(3) Diastereomers (4) Identical

COI063

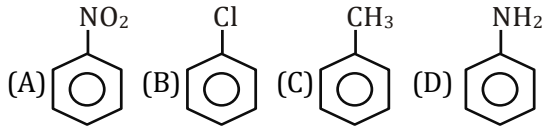
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**1**	**2**	**3**	**3**	**4**	**1**	**3**	**1**	**1**	**1**	**1**	**1**	**2**	**2**	**2**
Question	16	17	18	19	20	21	22	23	24	25					
Answer	**2**	**3**	**1**	**4**	**3**	**3**	**4**	**3**	**4**	**3**					

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Hydrocarbons

1. Arrange the following compound in reactivity order towards EAS (Electrophilic aromatic substitution) reaction ?



(1) A > B > C > D (2) B > C > A > D
(3) D > C > B > A (4) C > B > D > A

CHC228

2. Arrange the following in correct order of reactivity towards $Cl_2/h\upsilon$ –

(A) CH_4 (B) CH_3CH_3

(C) $CH_3CH_2CH_3$ (D) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(1) A > B > C > D (2) D > C > B > A
(3) B > C > A > D (4) C > B > D > A

CHC229

3. Reactivity order of the following alkenes for hydrochlorination (HCl) is

$CH_2 = CH - Cl$ (I) $(CH_3)_2C = CH_2$ (II)

$CH_2 = CH - CHO$ (III)

(1) I > II > III (2) II > III > I
(3) II > I > III (4) III > I > II

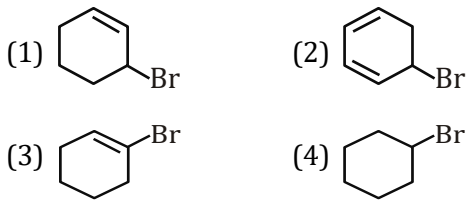
CHC230

4. Arrange the following compounds in decreasing order of reactivity for EAR :-

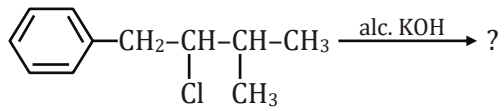
(I) $CH_2=CH_2$ (II) $CH_3-CH=CH_2$
(III) $Ph_2C=CH-CH_3$ (IV) $CH_2=CH-CHO$
(1) IV > I > II > III (2) III > II > I > IV
(3) II > III > I > IV (4) II > III > IV > I

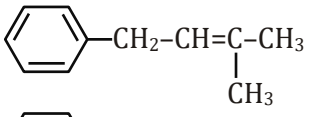
CHC231

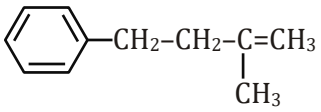
5. Which of the following is maximum reactive for dehydrohalogenation :–

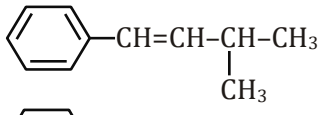


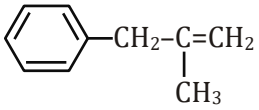
CHC232

6. 

(1) 

(2) 

(3) 

(4) 

CHC233

7. The decreasing order of bond dissociation energy of C–H bond in given compound is :-

(A) CH_3-H (B) CH_3-CH_2-H

(C) $CH_2=CH-CH_2-H$ (D) 

(1) A > B > C > D (2) D > A > B > C
(3) D > C > B > A (4) C > B > A > D

CHC234

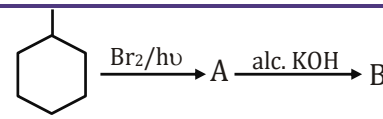
8. Match column-I and column-II and select the answer :

Column-I	Column-II
(a) $CH_3-CH=CH_2 \xrightarrow{?} CH_3-\underset{\underset{Br}{\|}}{CH}-CH_3$	(p)HBr with R_2O_2
(b) $CH_3-CH=CH_2 \xrightarrow{?} CH_2(Br)-CH=CH_2$	(q) Br_2/CCl_4
(c) $CH_3-CH=CH_2 \xrightarrow{?} CH_3-\underset{\underset{Br}{\|}}{CH}-CH_2Br$	(r) NBS
(d) $CH_3-CH=CH_2 \xrightarrow{?} CH_3-CH_2-\underset{\underset{Br}{\|}}{CH_2}$	(s) HBr/CCl_4

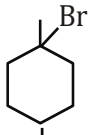
(1) (a-s), (b-r), (c-q), (d-p)
(2) (a-p), (b-r), (c-q), (d-s)
(3) (a-s), (b-q), (c-r), (d-p)
(4) (a-q), (b-r), (c-s), (d-p)

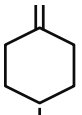
CHC235

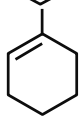
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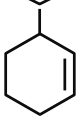
9. 

Major B is :-

(1) 

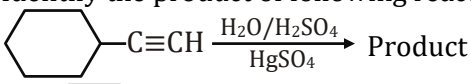
(2) 

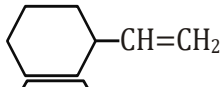
(3) 

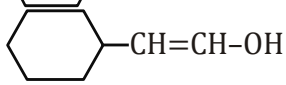
(4) 

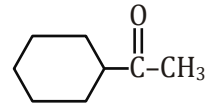
CHC236

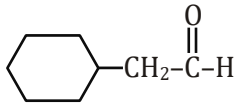
10. Identify the product of following reaction ?



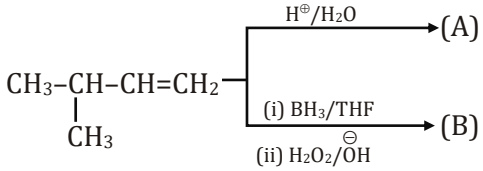
(1) 

(2) 

(3) 

(4) 

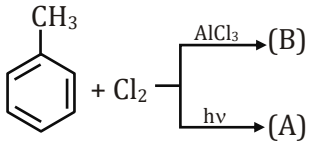
CHC237

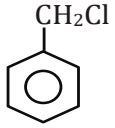
11. 

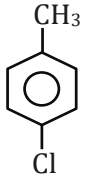
A and B are :-
(1) Chain isomers
(2) Functional isomers
(3) Metamers
(4) Position isomers

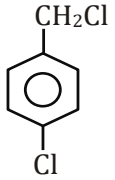
CHC238

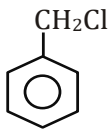
12. A and B respectively are :-

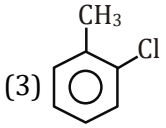


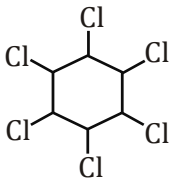
(1) 

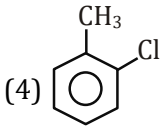
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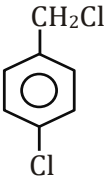
(2) 

,


(3) 

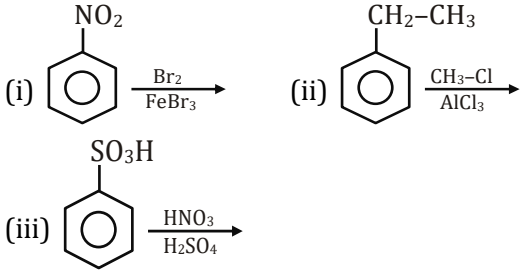
,


(4) 

,


CHC239

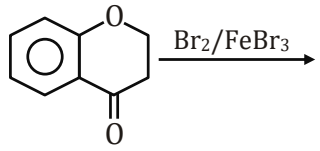
13. What are the major product of following reaction ?



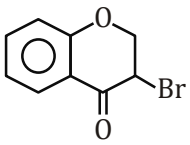
CHC240

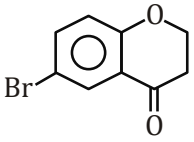
14. Which of the following yield But-2-ene on dehydration with con. H_2SO_4 ?
(1) 2-Methyl-2-butanol
(2) 2-Propanol
(3) 2-Methyl-2-propanol
(4) Butan-2-ol

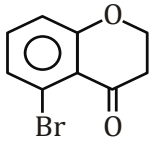
CHC241

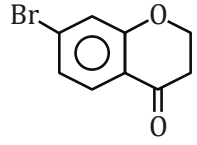
15. 

A (Major) is

(1) 

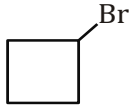
(2) 

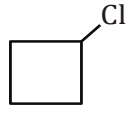
(3) 

(4) 

CHC242

16. 1–Bromo–3–chloro cyclobutane on reaction with 2–equivalent of sodium in ether gives

(1) 

(2) 

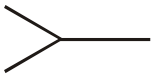
(3) 

(4) 

CHC243

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17. Which of the following alkane is synthesised from single alkyl halide by wurtz reaction:-

(1)  (2) 

(3)  (4) 

CHC244

18. The compound which produce propane on heating with HI in presence of red P is

(a) $CH_3CH_2CH_2I$ (b) CH_3CH_2CHO

(c) $CH_3CH_2CH_2OH$ (d) CH_3CH_2COOH

(1) Only c (2) b, c

(3) b, c, d (4) All

CHC245

19. The compound that decolourises alkaline $KMnO_4$ is :–

(1) C_6H_6 (Benzene) (2) C_6H_{12}

(3) C_6H_{14} (4) CCl_4

CHC246

20. Identify the hydrocarbon which on ozonolysis gives :-

(a) Only acetone

(b) 1 equivalent of propanal + 1 equivalent of formaldehyde

(c) 1 equivalent of acetone + 1 equivalent of acetaldehyde

CHC247

21. The reaction that give good yield of n-Hexane is :-

(1) $CH_3CH_2CH_2–Cl + CH_3CH_2CH_2Cl \xrightarrow{Na/dry\ ether}$

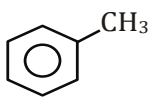
(2) $CH_3CH_2Cl + CH_3CH_2CH_2CH_2Cl \xrightarrow{Na/dry\ ether}$

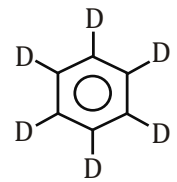
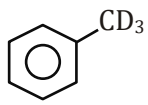
(3) $CH_3–Cl + CH_3CH_2CH_2CH_2CH_2Cl \xrightarrow{Na/dry\ ether}$

(4) All of them

CHC248

22. $CaC_2 \xrightarrow{D_2O} A \xrightarrow[Cu\ tube]{Red\ hot} B$; B is

(1)  (2) 

(3)  (4) 

CHC249

23. Which of the following statement is correct for the given compounds ?

(a) $CH_3 – CH = CH_2$ (b) $CH_3 – C \equiv CH$

(1) (a) is more reactive than (b) for electrophilic addition reaction.

(2) Both (a) and (b) reacts with cold alkaline $KMnO_4$

(3) (b) can be converted into (a) by reacting with H_2—Pd/$BaSO_4$.

(4) All

CHC250

24. Birch reduction is observed in

(1) $CH_3–CH=CH–CH_3$

(2) $HC\equiv CH$

(3) $H_3C–C\equiv CH$

(4) $H_3C–C\equiv C–CH_3$

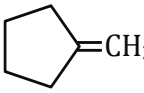
CHC251

25. Give products of the following reactions :-

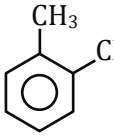
(a) $CH_2=CH_2 \xrightarrow[(ii)\,Zn/H_2O]{(i)\,O_3}$

(b) $CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2 \xrightarrow[(ii)\,Zn/H_2O]{(i)\,O_3}$

(c)  $\xrightarrow[(ii)\ Zn,\ H_2O]{(i)\ O_3}$

(d)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

(e)  $\xrightarrow[\Delta]{Ozonolysis}$

(f)  $\xrightarrow[\Delta]{Ozonolysis}$

CHC252

26. A $\xrightarrow{Kolbe's\ electrolysis}$ propyne will be

(1) $CH_3–CH_2–CH_2–COOK$

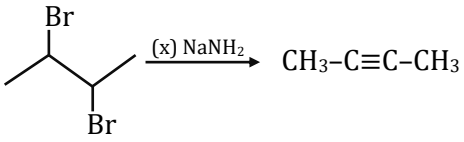
(2) $\underset{\displaystyle COOK}{\overset{\displaystyle |}{CH_2}}—\underset{\displaystyle COOK}{\overset{\displaystyle |}{CH}}—CH_3$

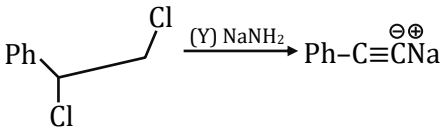
(3) $\underset{\displaystyle COOK}{\overset{\displaystyle |}{CH}} = \underset{\displaystyle COOK}{\overset{\displaystyle |}{C}}—CH_3$

(4) $\underset{\displaystyle COOK}{\overset{\displaystyle |}{CH_2}}—CH=CH_2$

CHC253

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27. (a) 

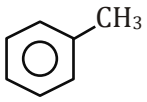
(b) 

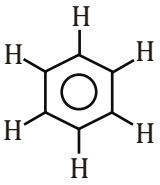
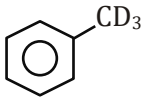
If X and Y are moles used then sum of [X + Y =] is :-

(1) 3 (2) 4 (3) 2 (4) 5

CHC254

28. $CaC_2 \xrightarrow{H_2O} A \xrightarrow[\text{Cu tube}]{\text{Red hot}} B$; B is

(1)  (2) 

(3)  (4) 

CHC255

29. Give products of the following reactions :-

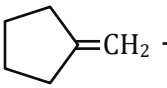
(a) $CH_2=CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

(b) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

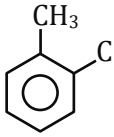
(c) $CH_3-C\equiv C-CH_3 \xrightarrow[(ii)\ H_2O]{(i)\ O_3}$

(d)  $\xrightarrow[(ii)\ H_2O]{(i)\ O_3}$

(e) $CH_3-CH=C=CH-CH_3 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

(f)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

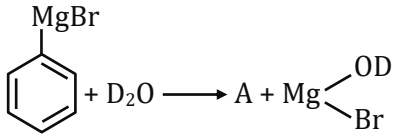
(g)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

(h)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

(i) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2 \xrightarrow[\Delta]{KMnO_4}$

(j) $CH_3-CH=CH-CH_3 \xrightarrow[\Delta]{KMnO_4}$

CHC256

30. 

The compound Ⓐ is :-

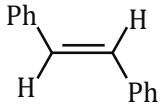
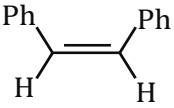
(1) Benzene (2) Duetero benzene

(3) Duetero toluene (4) Both (2) and (3)

CHC257

31. $Ph-CH=CH-Ph \xrightarrow[CCl_4]{Cl_2} X \xrightarrow{2NaNH_2} Y \xrightarrow[BaSO_4]{H_2+Pd} Z$

Product Z is

(1)  (2) 

(3) $Ph-\underset{\displaystyle Ph}{\overset{\displaystyle |}{C}}=CH_2$ (4) $Ph-C\equiv C-Ph$

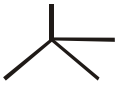
CHC258

32. Among the following, the compound which has highest boiling point is :-

(1) 

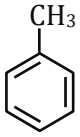
(2) 

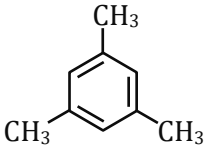
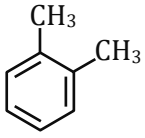
(3) 

(4) 

CHC259

33. $3CH_3-C\equiv CH \xrightarrow[\text{Cu tube}]{\text{Red hot}} ?$

(1)  (2) 

(3)  (4) 

CHC260

34. Reactivity order of the following compounds for hydrochlorination (HCl) is

(I) $CH_2 = CH - Cl$ (II) $(CH_3)_2C = CH_2$

(III) $CH_2 = CH - CHO$

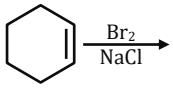
(1) I > II > III (2) II > III > I

(3) II > I > III (4) III > I > II

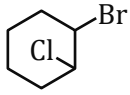
CHC261

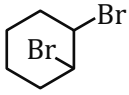
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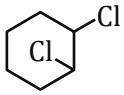
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35.  ?

Which of the following product will not be obtained :-

(1) 

(2) 

(3) 

(4) Both (1) and (2)

CHC262

36. $CH_2=CH-CCl_3 \xrightarrow{HBr} ?$

(1) $CH_2=CH-CBr_3$

(2) $CH_2(Br)-CH_2-CCl_3$

(3) $CH_3-CH(Br)-CCl_3$

(4) $CH_2(Br)-CH(Br)-CCl_3$

CHC263

37. Select the chain propagation steps in the free radical chlorination of methane :-

(a) $Cl_2 \longrightarrow 2\dot{C}l$

(b) $\dot{C}l + CH_4 \longrightarrow CH_3Cl + \dot{H}$

(c) $\dot{C}l + CH_4 \longrightarrow \dot{C}H_3 + HCl$

(d) $\dot{H} + Cl_2 \longrightarrow HCl + \dot{C}l$

(e) $\dot{C}H_3 + Cl_2 \longrightarrow CH_3-Cl + \dot{C}l$

(1) b, c, e (2) a, c, e (3) c, e (4) b, c, d

CHC264

38. A hydrocarbon X (V.D. = 35) forms only one monochloro substitution product, X is :-

(1) Neopentane (2) Cyclopentane

(3) Pent-1-ene (4) n-pentane

CHC265

39. Toluene reacts with 1 mol of Cl_2 in the presence of light to give :–

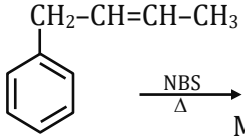
(1) Benzyl chloride

(2) Benzoyl chloride

(3) p–chlorotoluene

(4) o–chlorotoluene

CHC266

40.  $\xrightarrow[\Delta]{NBS}$ (A) Major

Compound (A) is :

(1) $Ph-CH_2-CH=CH-CH_2Br$ (Br)

(2) $Ph-CH(Br)-CH=CH-CH_3$

(3) $Ph-CH_2-CH_2-CH(Br)-CH_3$

(4) $Ph-CH_2-CH(Br)-CH_2-CH_3$

CHC267

41. Match column-I and column-II and select the answer :

Column-I		Column-II	
(a)	$CH_3-CH=CH_2 \xrightarrow{?} CH_3-CH(Br)-CH_3$	(p)	HBr followed by hydrolysis
(b)	$CH_3-CH=CH_2 \xrightarrow{?} CH_2(Br)-CH=CH_2$	(q)	Br_2/CCl_4
(c)	$CH_3-CH=CH_2 \xrightarrow{?} CH_3-CH(Br)-CH_2(Br)$	(r)	NBS
(d)	$CH_3-CH=CH_2 \xrightarrow{?} CH_3-CH(OH)-CH_3$	(s)	HBr

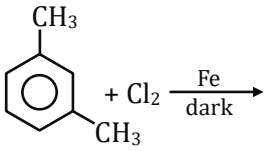
(1) (a-s), (b-r), (c-q), (d-p)

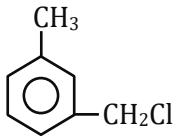
(2) (a-p), (b-r), (c-q), (d-s)

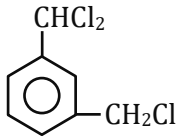
(3) (a-s), (b-q), (c-r), (d-p)

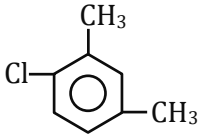
(4) (a-q), (b-r), (c-s), (d-p)

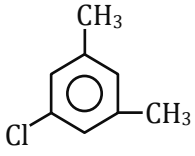
CHC268

42.  + $Cl_2 \xrightarrow[dark]{Fe}$ A ; Compound (A) is :

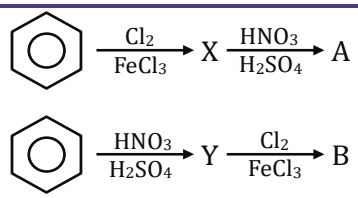
(1) 

(2) 

(3) 

(4) 

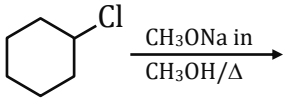
CHC269

43. 

A and B are :-
(1) Identical structure
(2) Position isomers
(3) Geometrical isomers
(4) None of the above

CHC270

44. Given reaction mainly proceeds through –



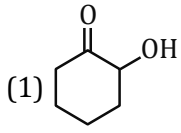
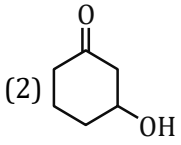
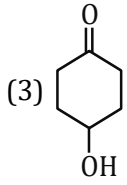
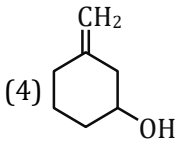
(1) E_1 (2) S_N1 (3) S_N2 (4) E_2

CHC271

45. In E_2 elimination some compounds follow Hoffmann rule which means
(1) The double bond goes to the most substituted position
(2) The compound is resistant to elimination
(3) No double bond is formed
(4) The double bond goes mainly toward the least substituted carbon.

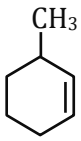
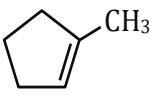
CHC272

46. Maximum dehydration takes place that of -

(1) 
(2) 
(3) 
(4) 

CHC273

47. Which is maximum reactive towards hydrogenation ?

(1) 
(2) 
(3) 
(4) 

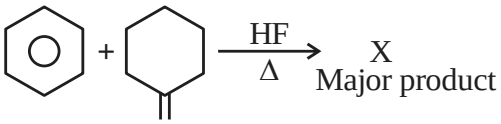
CHC274

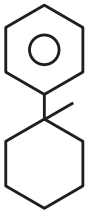
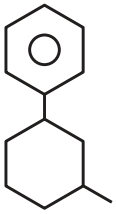
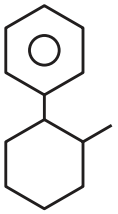
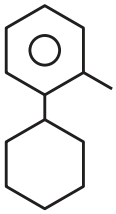
48. $CH_3-C\equiv CH + \overset{\oplus}{Na}\overset{\ominus}{N}H_2 \rightarrow A \xrightarrow{CH_3-Cl} B$

Product B is-
(1) $CH_3-C\equiv C-CH_3$ (2) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2$
(3) $CH_3-CH_2-C\equiv CH$ (4) $CH_3-C\equiv C-Cl$

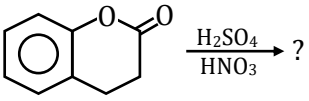
CHC275

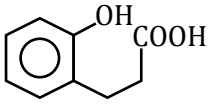
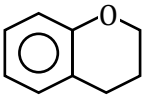
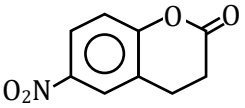
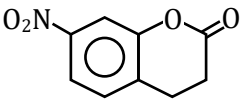
49. 'X' is :



(1) 
(2) 
(3) 
(4) 

CHC276

50. 

(1) 
(2) 
(3) 
(4) 

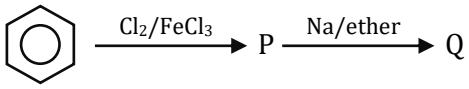
CHC277

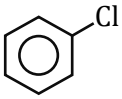
51. The compounds 1-butyne and 2-butyne can be distinguished by using
(1) Bromine water (2) alk. $KMnO_4$ (cold)
(3) Tollens's reagent (4) Chlorine gas

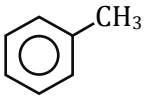
CHC278

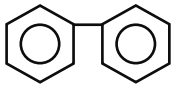
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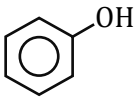
52. The end product (Q) in the following sequence of reactions is



(1) 

(2) 

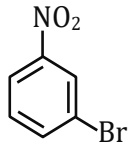
(3) 

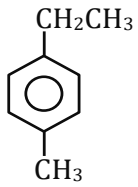
(4) 

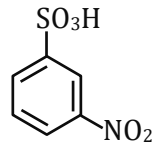
CHC279

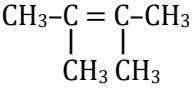
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	14	15	16
Answer	**3**	**2**	**3**	**2**	**2**	**3**	**2**	**1**	**3**	**3**	**4**	**1**	**4**	**2**	**3**
Question	17	18	19	21	22	23	24	26	27	28	30	31	32	33	34
Answer	**4**	**4**	**2**	**1**	**3**	**4**	**4**	**3**	**4**	**3**	**2**	**2**	**3**	**3**	**3**
Question	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49
Answer	**3**	**2**	**3**	**2**	**1**	**2**	**1**	**3**	**2**	**4**	**4**	**2**	**3**	**1**	**1**
Question	50	51	52												
Answer	**3**	**3**	**3**												

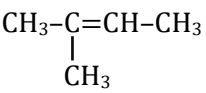
13. (i) 

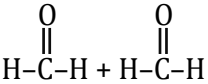
(ii) 

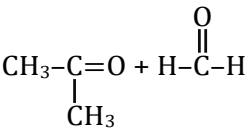
(iii) 

20. (a) 

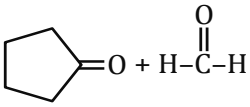
(b) $CH_3CH_2CH{=}CH_2$

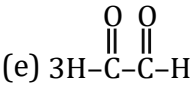
(c) 

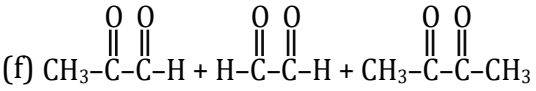
25. (a) 

(b) 

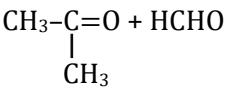
(c) 

(d) 

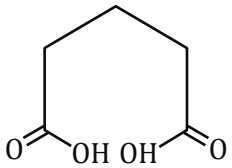
(e) 

(f) 

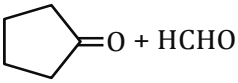
29. (a) 2HCHO

(b) 

(c) 

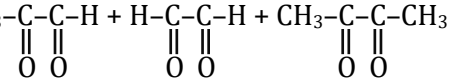
(d) 

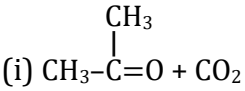
(e) $2CH_3 - CHO + CO_2$

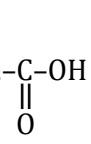
(f) 

(g) 

(Glyoxal)

(h) 

(i) 

(j) 

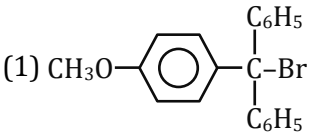
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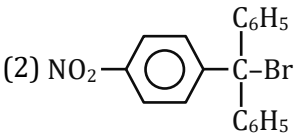
Haloalkanes and Haloarenes

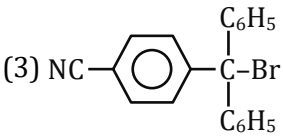
1. The order of reactivity of the following alkyl halides for a S_N2 reaction is -
(1) RF > RCl > R –Br > R–I
(2) R–F > R –Br > R–Cl > R–I
(3) R–Cl > R –Br > R–F > R–I
(4) R–I > R –Br > R–Cl > R–F

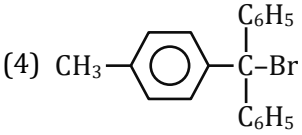
CHH178

2. Which one of the following is most reactive for S_N1 reaction :

(1) 

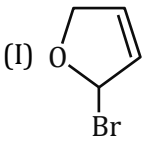
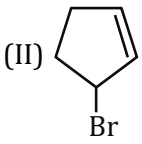
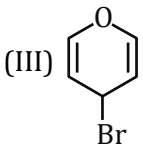
(2) 

(3) 

(4) 

CHH179

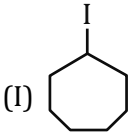
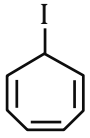
3. Among the bromides given below, the order of reactivity of S_N1 reaction is -

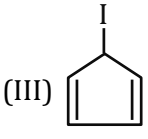
(I)  (II)  (III) 

(1) III > I > II (2) III > II > I
(3) II > III > I (4) II > I > III

CHH180

4. Compare rate of S_N1 :-

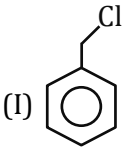
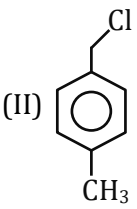
(I)  (II) 

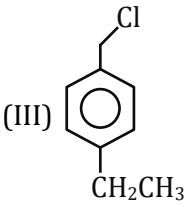
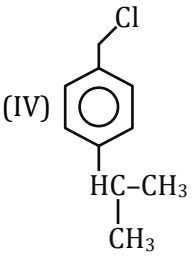
(III) 

(1) II > III > I (2) II > I > III
(3) I > II > III (4) III > II > I

CHH181

5. Reactivity order for S_N1 reaction

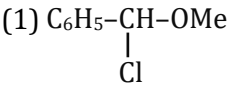
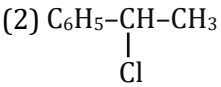
(I)  (II) 

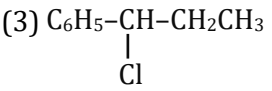
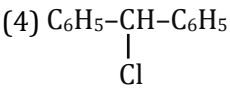
(III)  (IV) 

(1) I > II > III > IV (2) II > III > IV > I
(3) IV > III > II > I (4) III > IV > II > I

CHH182

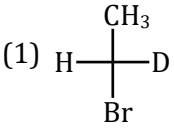
6. Which of the following undergoes fastest reaction with aqueous NaOH solution

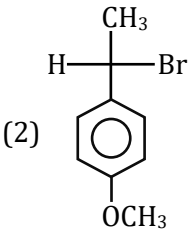
(1)  (2) 

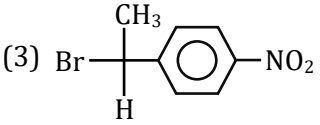
(3)  (4) 

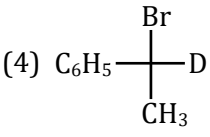
CHH183

7. Under identical conditions, solvolysis of which of the following substrate would lead to maximum racemisation :-

(1) 

(2) 

(3) 

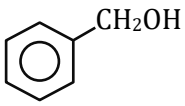
(4) 

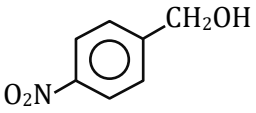
CHH184

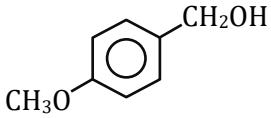
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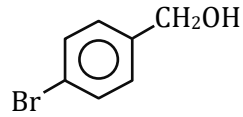
AIM : FREE EDUCATION TO ALL APUL

8. Which of the following alcohols is most reactive towards substitution with HBr ?

(1) 

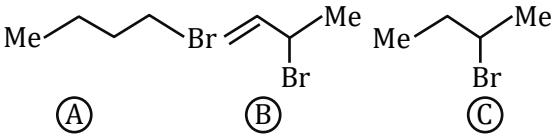
(2) 

(3) 

(4) 

CHH185

9. Consider the following bromides :-



The correct order of S_N1 reactivity is

(1) A > B > C (2) B > C > A

(3) B > A > C (4) C > B > A

CHH186

10. Of the following statements which are true for S_N1 reaction.

(a) Tertiary alkyl halides react faster than secondary.

(b) The absolute configuration of the product is opposite to that of the reactant when an optical active substrate is used.

(c) The reaction shows first order kinetics.

(d) The rate of reaction depends markedly on the nucleophilicity of the nucleophile.

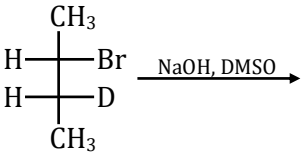
(e) The mechanism is two step.

(f) Carbocations are intermediate.

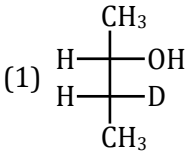
(g) Rate $\propto$ [Alkyl halides]

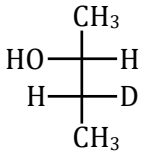
(h) The rate of the reaction depends on the nature of the leaving group.

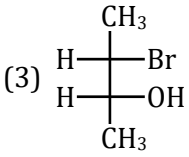
CHH187

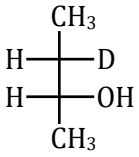
11.  Product.

Product .....is :-

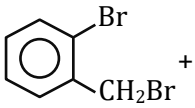
(1) 

(2) 

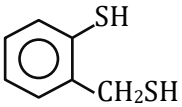
(3) 

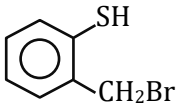
(4) 

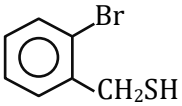
CHH188

12.  + KSH ⟶ A.

Compound Ⓐ is

(1) 

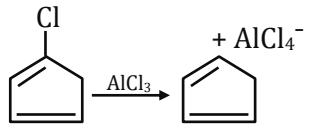
(2) 

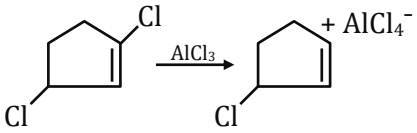
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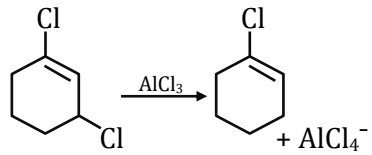
(4) 

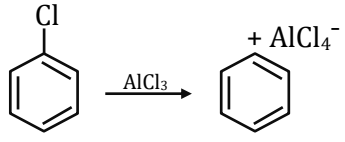
CHH189

13. Which of the following reaction is feasible ?

(1) 

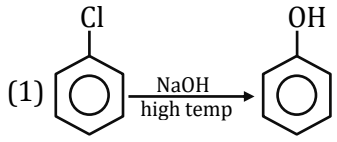
(2) 

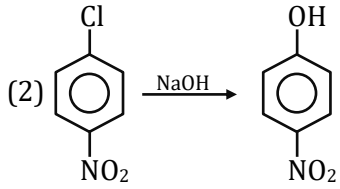
(3) 

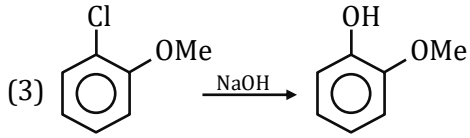
(4) 

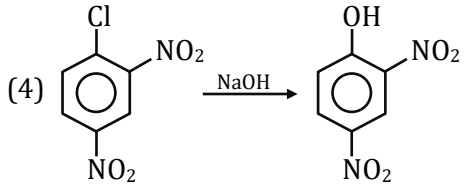
CHH190

14. Which reaction is least feasible

(1) 

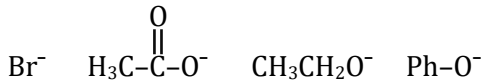
(2) 

(3) 

(4) 

CHH191

15. Arrange the following in order of their leaving group tendency?



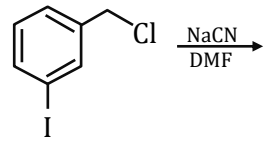
(I) (II) (III) (IV)

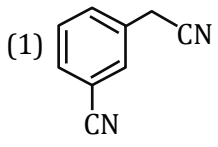
(1) I > II > III > IV (2) I >III > II > IV

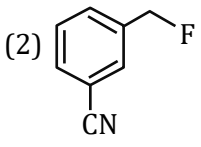
(3) I > II > IV > III (4) I > IV > III > II

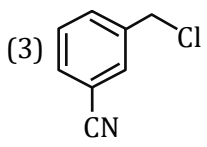
CHH192

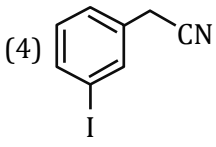
16. The structure of the major product formed in the following reaction is :



(1) 

(2) 

(3) 

(4) 

CHH193

17. Which of the following on heating with aqueous KOH, produces acetaldehyde?

(1) CH_2ClCH_2Cl (2) CH_3CHCl_2

(3) CH_3COCl (4) CH_3CH_2Cl

CHH194

18. $R–OH \xrightarrow{\text{Reagent}} R–Cl$

Which of the following reagents can be used for this reaction.

(a) $HCl/ZnCl_2$ (b) NaCl

(c) PCl_3 (d) PCl_5

(e) $SOCl_2$

(1) a, c, d, e (2) a, b, c, d

(3) a, b, d, e (4) All of the above

CHH195

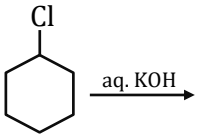
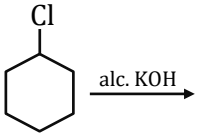
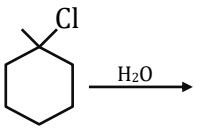
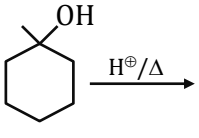
19. $R–OH \xrightarrow[\text{Pyridine}]{SOCl_2} R–Cl + SO_2 + HCl$

This reaction follows which of the following mechanisms ?

(1) S_N1 (2) S_N2 (3) E_1 (4) E_2

CHH196

20. Match the column-I and column-II and select the correct answer :

Column-I		Column-II	
(a)		(p)	E_1
(b)		(q)	S_N1
(c)		(r)	S_N2
(d)		(s)	E_2

(1) (a-s); (b-r); (c-p); (d-q)

(2) (a-r); (b-s); (c-q); (d-p)

(3) (a-r); (b-s); (c-p); (d-q)

(4) (a-q); (b-p); (c-r); (d-s)

CHH197

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21. Identify the set of reagents/reaction conditions 'X' and 'Y' in the following set of transformations :-

$CH_3-CH_2-CH_2Br \xrightarrow{X}$ Product $\xrightarrow{Y}$ $CH_3-\underset{\underset{Br}{|}}{CH}-CH_3$

(1) X = dilute aqueous NaOH, 20°C,
Y = Br_2/acetic acid, 20°C

(2) X = concentrated alcoholic NaOH, 80°C;
Y = HBr/$CHCl_3$, 20°C

(3) X = dilute aqueous NaOH 20°,
Y = $Br_2/CHCl_3$, 0°C

(4) X = concentrated alcoholic NaOH, 80°C;
Y = $Br_2/CHCl_3$, 0°C

CHH198

22. Which of the following statement is true for the major product obtained in the given reaction :-

$CH_4 + \underset{\text{(Excess)}}{Cl_2} \xrightarrow{h\nu}$ (major product) :-

(1) Reacts with $AgNO_3$(aq.) to give white ppt of AgCl
(2) Is used as an aneasthetic
(3) Has zero dipole moment
(4) All

CHH199

23. Which of the following is a feasible substitution ?

(1) R–Br $\xrightarrow{Hg_2F_2}$ (2) R–Br $\xrightarrow[\text{acetone}]{NaI}$
(3) R–Cl $\xrightarrow{AgF}$ (4) All of these

CHH200

24. Which is best Nucleophile ?

(1) $CH_3-\overset{\ominus}{C}H_2$ (2) $CH_3-CH_2-\overset{\ominus}{C}H_2$
(3) $CH_3-\overset{\ominus}{N}H$ (4) $\overset{\ominus}{O}H$

CHH201

25. Correct statement about S_N2 reaction is

(1) (a) $CH_3-Cl + \overset{\ominus}{O}H \rightarrow$
(b) $CH_3-Cl + \overset{\ominus}{S}H \rightarrow$
S_N2 order → a > b

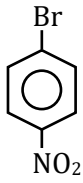
(2) (a) $CH_3-CH_2-Cl + \overset{\ominus}{O}H \rightarrow$
(b) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3 + \overset{\ominus}{O}H \rightarrow$
S_N2 order → b > a

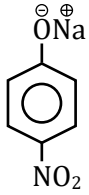
(3) (a) $CH_3-CH_2-Cl + \overset{\ominus}{O}H \rightarrow$
(b) $CH_3-CH_2-Br + \overset{\ominus}{O}H \rightarrow$
S_N2 order → b > a

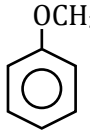
(4) (a) $CH_3-\underset{\underset{O}{\|}}{C}-CH_2-Cl \rightarrow$
(b) $CH_3-CH_2-CH_2 + Cl \rightarrow$
S_N2 order → b > a

CHH202

26. Which is appropriate set of reactants for the preparation of 1-Methoxy-4-nitrobenzene

(1)  + $CH_3-\overset{\ominus}{O}\overset{\oplus}{Na}$

(2)  + CH_3-Br

(3)  $\xrightarrow[HNO_3]{H_2SO_4}$

(4) (2) and (3) Both

CHH203

27. Given below are two statements:
Statement-I : Ethyne is formed from dibromoethane by using alcoholic KOH followed by $NaNH_2$.
Statement-II : Ethyne is formed from geminal dibromo ethane by using one equivalent of alcoholic KOH.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement-I and Statement-II both are correct.
(2) Statement-I and II both are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

CHH204

ANSWER KEY

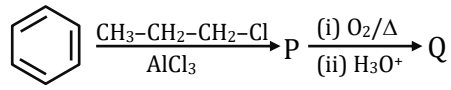
Question	1	2	3	4	5	6	7	8	9	10	11
Answer	**4**	**1**	**1**	**2**	**2**	**1**	**2**	**3**	**2**	**a-T,b-F,c-T,d-F,e-T,f-T,g-T,h-T**	**2**

Question	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Answer	**3**	**3**	**3**	**3**	**4**	**2**	**1**	**2**	**2**	**2**	**3**	**4**	**2**	**3**	**4**

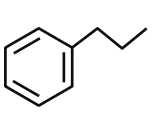
Question	27
Answer	**3**

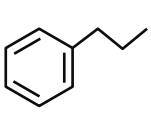
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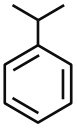
Alcohols, Phenols and Ethers

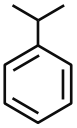
1. 

Q + phenol, P & Q are -

(1)  and $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$

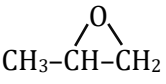
(2)  and $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

(3)  and $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

(4)  and $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-H$

CAE140

2. $CH_3-\underset{OH}{CH}-\underset{Cl}{CH_2} + Na \rightarrow A \xrightarrow[2.\ H_3O^+]{1.\ CH_3MgCl} B$, B is :

(1) 

(2) $CH_3-\underset{OH}{CH}-CH_2-CH_3$

(3) $CH_3-\underset{CH_3}{CH}-CH_2-OH$

(4) $CH_3-\underset{OH}{CH}-\underset{OH}{CH_2}$

CAE141

3. Match the following :

Column-I		Column-II	
(a)	Phenol + Neutral $FeCl_3$	(p)	No reaction
(b)	Phenol + Br_2(aq.)	(q)	Violet colour
(c)	Phenol + $NaHCO_3$	(r)	White ppt.
(d)	Picric acid + $NaHCO_3$	(s)	CO_2 gas is evolved

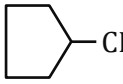
(1) a-q, b-r, c-p, d-s (2) a-r, b-q, c-p, d-s
(3) a-q, b-p, c-r, d-s (4) a-q, b-r, c-s, d-p

CAE142

4. Which reaction is related to Williamson ether synthesis ?

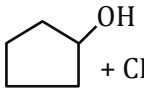
(1) $CH_3-CH_2-CH_2-Cl + (CH_3)_3C-\overset{-}{O}\overset{+}{N}a \rightarrow$

(2) $CH_3-\underset{CH_3}{CH}-Cl + CH_3-CH_2-\overset{-}{O}\overset{+}{N}a \rightarrow$

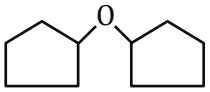
(3)  $CH_2-Cl + CH_3 + CH_3-\overset{-}{O}\overset{+}{N}a$

(4) $Ph-CH_2-Cl + CH_3-\overset{-}{O}\overset{+}{N}a \rightarrow$

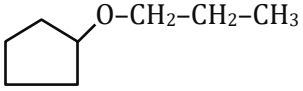
CAE143

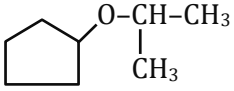
5.  $+ CH_3-\underset{OH}{CH}-CH_3 \xrightarrow[140°C]{H^\oplus}$ Products

Which is not obtained in product ?

(1) 

(2) $CH_3-\underset{CH_3}{CH}-O-\underset{CH_3}{CH}-CH_3$

(3) 

(4) 

CAE144

6. $CH_3-\underset{CH_3}{CH}-CH_2-O-CH_2-Ph \xrightarrow{H-I} ?$

(1) $CH_3-\underset{CH_3}{CH}-CH_2-I + HO-CH_2-Ph$

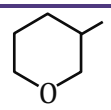
(2) $CH_3-\underset{CH_3}{\overset{I}{C}}-CH_3 + HO-CH_2-Ph$

(3) $CH_3-\underset{CH_3}{CH}-CH_2-OH + I-CH_2-Ph$

(4) $CH_3-\underset{CH_3}{\overset{OH}{C}}-CH_3 + I-CH_2-Ph$

CAE145

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7.  + H–Br ⟶ ?

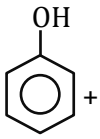
(1) $Br-(CH_2)_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-OH$

(2) $HO-(CH_2)_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Br$

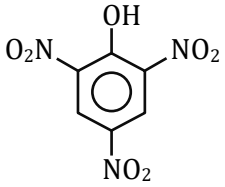
(3) $HO-(CH_2)_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_3$

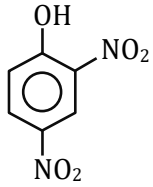
(4) $Br-(CH_2)_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

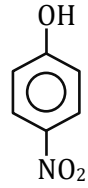
CAE146

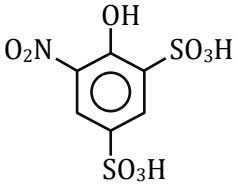
8.  + Conc. H_2SO_4 ⟶ A $\xrightarrow{\text{Conc. } HNO_3}$ B

Product 'B' is ?

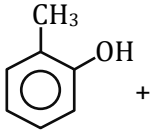
(1) 

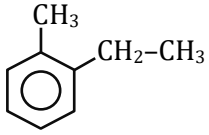
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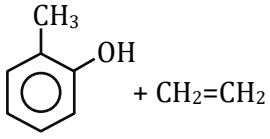
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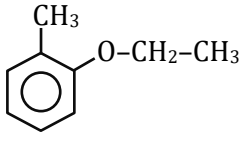
(4) 

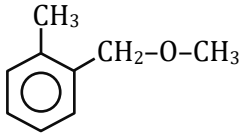
CAE147

9.  + Na ⟶ A $\xrightarrow{CH_3-CH_2-Cl}$ B ?

(1) 

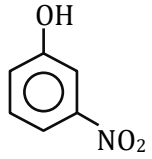
(2) 

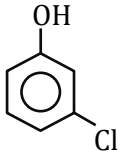
(3) 

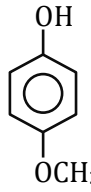
(4) 

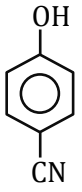
CAE148

10. Which is most acidic ?

(1) 

(2) 

(3) 

(4) 

CAE149

11. Ethanol and Phenol can be differentiated by :-

(1) Br_2/H_2O

(2) O_2

(3) NaOH

(4) (1) and (3) both

CAE150

12. Which is best method to obtained tert butyl methyl ether -

(1) $CH_3-OH + CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH \xrightarrow[140°C]{H^{\oplus}}$

(2) $CH_3-Cl + (CH_3)_3\left(-\overset{-}{O}\overset{+}{N}a\right) \rightarrow$

(3) $(CH_3)_3C-Cl + CH_3-\overset{-}{O}\overset{+}{N}a \rightarrow$

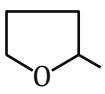
(4) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Cl + CH_3-\overset{-}{O}\overset{+}{N}a \rightarrow$

CAE151

13. Which gives ether as a major product ?

(1) $Ph-Cl + CH_3-\overset{-}{O}\overset{+}{N}a \rightarrow$

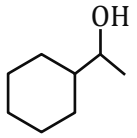
(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Cl + CH_3-\overset{-}{O}\overset{+}{N}a \rightarrow$

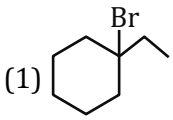
(3)  + $CH_3-CH_2-\overset{-}{O}\overset{+}{N}a \rightarrow$

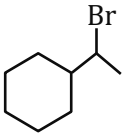
(4) $CH_2=CH-Cl + Ph-\overset{-}{O}\overset{+}{N}a \rightarrow$

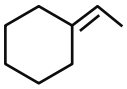
CAE152

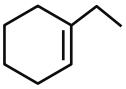
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14.  + H–Br ⟶ ?

(1) 

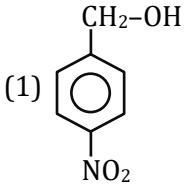
(2) 

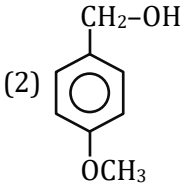
(3) 

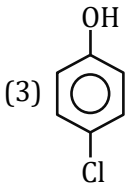
(4) 

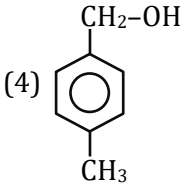
CAE153

15. Which is maximum reactive towards Lucas reagent?

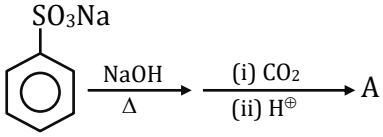
(1) 

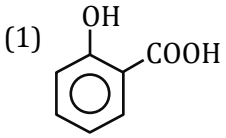
(2) 

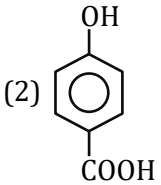
(3) 

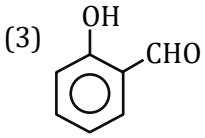
(4) 

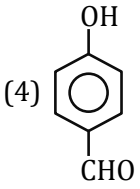
CAE154

16. 

(1) 

(2) 

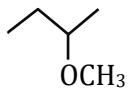
(3) 

(4) 

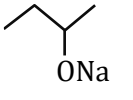
CAE155

17.  + $SOCl_2$ ⟶ A $\xrightarrow{CH_3-\overset{-}{O}\overset{+}{N}a}$ B ?

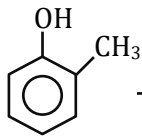
(1) 

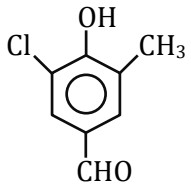
(2) 

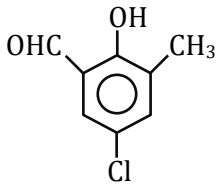
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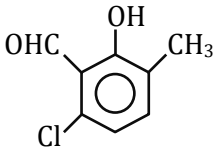
(4) 

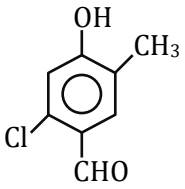
CAE156

18. 

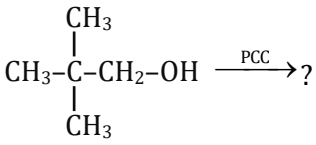
(1) 

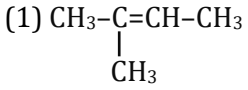
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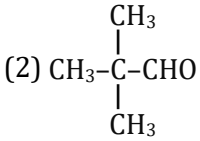
(3) 

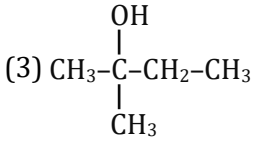
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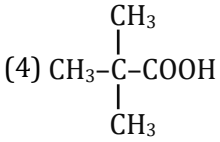
CAE157

19. 

(1) 

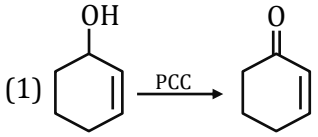
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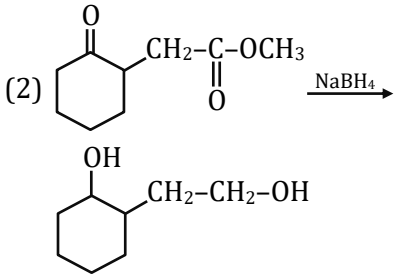
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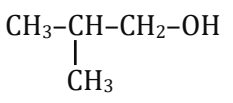
(4) 

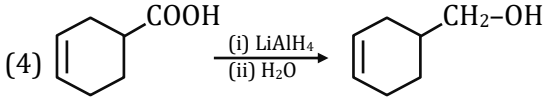
CAE158

20. Which is incorrect reaction ?

(1) 

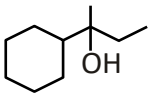
(2) 

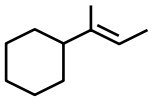
(3) H–CHO + $(CH_3)_2CHMgBr$ ⟶ 

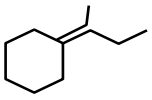
(4) 

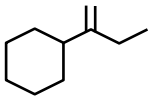
CAE159

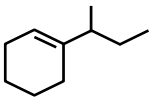
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21. Which of the following is not the product of dehydration of 

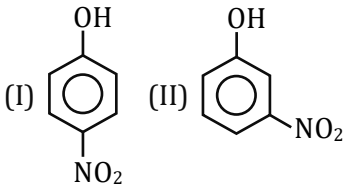
(1) 

(2) 

(3) 

(4) 

CAE160

22. 

Which of the following statements is correct about these phenols?

(1) –I effect of $-NO_2$ group on the –OH group is stronger in meta isomer (II) than in para isomer (I). Therefore, II is stronger acid than I.

(2) $-NO_2$ group has only –I effect on the –OH group in II, but it has –I as well as –M effect on the –OH group in I. Therefore, I is stronger acid than II.

(3) Conjugate base of II is more stable than that of I. Therefore, II is stronger acid than I.

(4) Both (1) and (3)

CAE161

ANSWER KEY

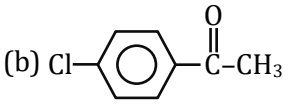
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**1**	**4**	**3**	**3**	**1**	**1**	**3**	**4**	**4**	**2**	**2**	**1**	**2**
Question	16	17	18	19	20	21	22								
Answer	**1**	**2**	**2**	**2**	**2**	**4**	**2**								

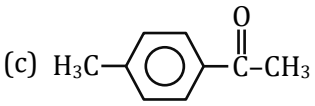
AIM : FREE EDUCATION TO ALL APUL

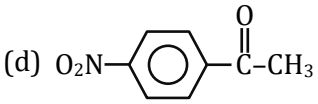
Aldehydes, Ketones and Carboxylic acids

1. Arrange the following compounds in decreasing order of Keq for nucleophilic addition reaction.

(a) Ph—CO—CH_3

(b) 

(c) 

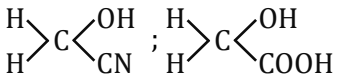
(d) 

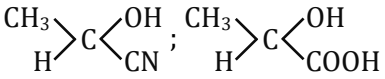
(1) a > b > c > d (2) c > a > b > d

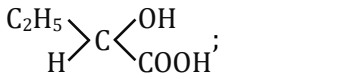
(3) d > b > a > c (4) d > a > b > c

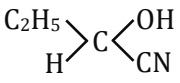
CAA194

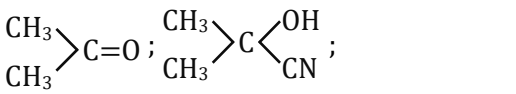
2. A carbonyl compound 'A' reacts with hydrogen cyanide to form a cyanohydrin 'B' which on hydrolysis gives an optically active alpha hydroxy acid 'C'. 'A' gives a positive iodoform test 'A', 'B' and 'C' are given by the set :-

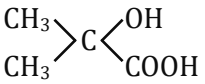
(1) HCHO; 

(2) CH_3CHO; 

(3) CH_3CH_2CHO; 

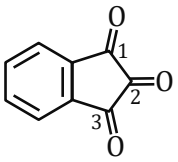


(4) 



CAA195

3. Which carbonyl group of the given compound is most reactive for hydration :-



(1) 3

(2) 1

(3) 2

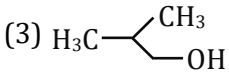
(4) All have equal reactivity

CAA196

4. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is-

(1) $CH_3CH_2CH(OH)CH_2CH_3$

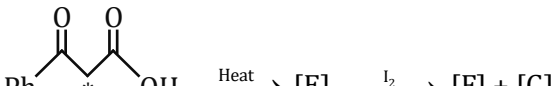
(2) $C_6H_5CH_2CH_2OH$

(3) 

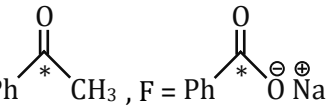
(4) $PhCHOHCH_3$

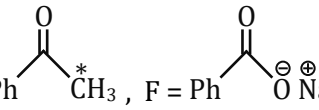
CAA197

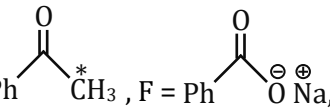
5. In the following reaction sequence, the correct structure of E, F and G are :

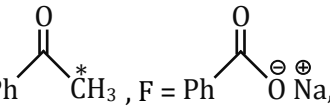


(* implies ^{13}C labelled carbon)

(1) E = 

(2) E = 

(3) E = 

(4) E = 

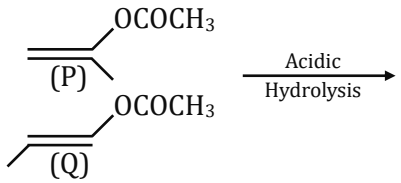
CAA198

6. Benzoyl chloride is prepared from benzoic acid by -

(1) Cl_2, hυ (2) SO_2Cl_2/hυ

(3) $SOCl_2$ (4) Cl_2, H_2O

CAA199

7. 

Product formed by (P) & (Q) can be differentiated by

(1) 2, 4 DNP

(2) Lucas reagent ($ZnCl_2$) conc. HCl

(3) $NaHSO_3$

(4) Fehlings solution

CAA200

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8. Compound A (molecular formula C_3H_8O) is treated with anhydrous CrO_3 to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammonical silver nitrate. B when treated with an aqueous solution of $H_2NCONHNH_2$, HCl and sodium acetate gives a product C. Identify the structure of C:-

(1) $CH_3CH_2CH{=}NNHCONH_2$

(2) $CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}{=}NNHCONH_2$

(3) $CH_3{-}\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}{=}NCONHNH_2$

(4) $CH_3CH_2CH{=}NCONHNH_2$

CAA201

9. In the following sequence of reactions

$CH_3CH_2OH \xrightarrow{P+I_2} A \xrightarrow[Ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$

then compound 'D' is -

(1) butanal (2) n–butyl alcohol

(3) n–propyl alcohol (4) propanal

CAA202

10. In Cannizzaro reaction given below :-

$2PhCHO \xrightarrow{:\overset{\ominus}{O}H} PhCH_2OH + PhCO_2^{\ominus}$

the slowest step is :-

(1) The abstraction of proton from the carboxylic group

(2) The deprotonation of $PhCH_2OH$

(3) The attack of $:\overset{\ominus}{O}H$ at the carboxyl group

(4) The transfer of hydride to the carbonyl group

CAA203

11. $CH_3CH_2COOH \xrightarrow[Red\ P]{Br_2} X \xrightarrow{NH_3} Y$

Y in the above reaction is -

(1) Lactic acid (2) Ethylamine

(3) Propylamine (4) Alanine

CAA204

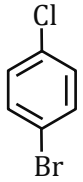
12. $R{-}\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}{-}OH + H\overset{(18)}{O}{-}CH_3 \xrightarrow{H^+} R{-}\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}{-}\overset{(18)}{O}{-}CH_3 + H_2O$

In above reaction $CH_3\overset{(18)}{O}H$ acts as :

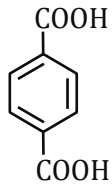
(1) Nucleophile (2) Base

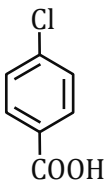
(3) Electrophile (4) Acid

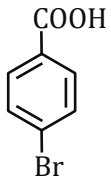
CAA205

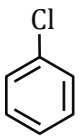
13. 

$+ Mg \underset{(1\ mole)}{\xrightarrow{THF}} A \xrightarrow{CO_2} B \xrightarrow{H^+} C$, C is :-

(1) 

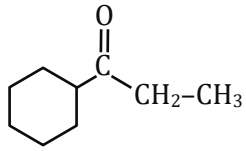
(2) 

(3) 

(4) 

CAA206

14. Which of the following compound can form hydrazone and will give iodoform test but will not give Tollen's test ?

(1) 

(2) $H_3C{-}\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}{-}CH_2{-}CH_3$

(3) $H_3C{-}\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}{-}CH_2{-}CHO$

(4) $CH_3{-}CH_2{-}OH$

CAA207

15. Which one of the following compounds will not be soluble in sodium bicarbonate ?

(1) Benzene sulphonic acid

(2) Benzoic acid

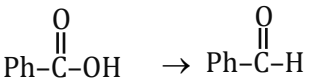
(3) o-Nitrophenol

(4) 2,4,6-Trinitrophenol

CAA208

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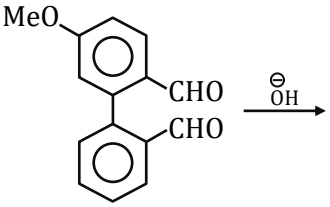
16. Find out correct sequence of reagent for following conversion :



(1) PCl_5, CH_3MgBr (2) CH_2N_2, $NaBH_4$
(3) $SOCl_2$, $H_2/Pd,BaSO_4$ (4) All of the above

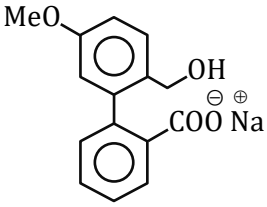
CAA209

17.

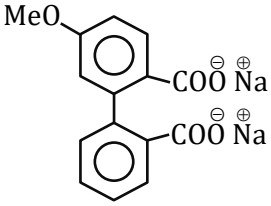


The major product formed in this reaction :-

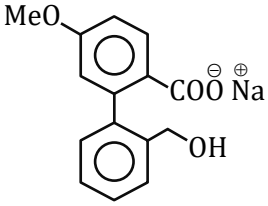
(1)



(2)



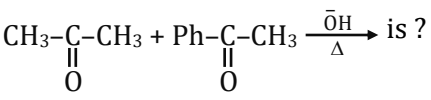
(3)



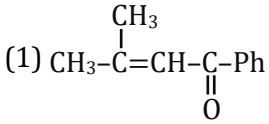
(4) None of these

CAA210

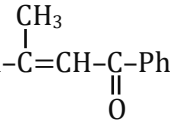
18. Cross aldol condensation product of



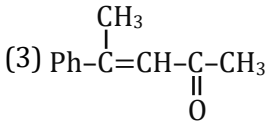
(1)



(2)



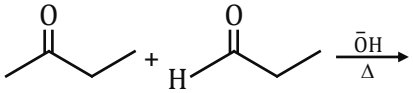
(3)



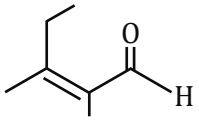
(4) 1 and 3 both

CAA211

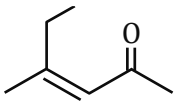
19. Which product is not obtained from the reaction below.



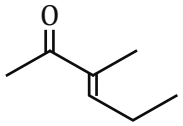
(1)



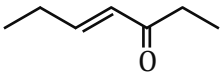
(2)



(3)



(4)



CAA212

20. $Ph\text{-}CHO + H\text{-}CHO \xrightarrow[(ii)\ H^{\oplus}]{(i)\ NaOH} A \xrightarrow[(ii)\ H_2O]{(i)\ 2CH_3MgCl} B$?

(1) $CH_3\text{–}\underset{\underset{OH}{|}}{CH}\text{–}CH_3 + Ph\text{–}CH_2\text{–}OH$

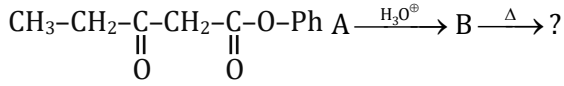
(2) $(CH_3)_3C\text{-}OH + Ph\text{-}CH_2\text{-}OH$

(3) $CH_3\text{-}CHO + Ph\text{-}CH_2\text{-}OH$

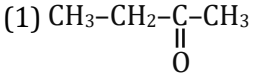
(4) $CH_3\text{-}CH_2\text{-}CH_2\text{-}OH + Ph\text{-}CH_2\text{-}OH$

CAA213

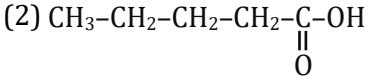
21.



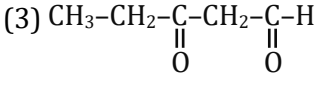
(1)



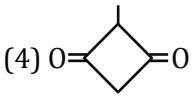
(2)



(3)

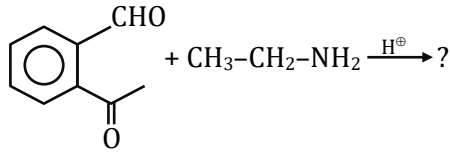


(4)

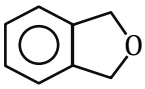


CAA214

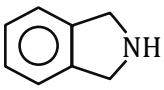
22.



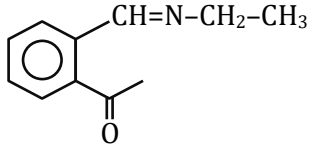
(1)



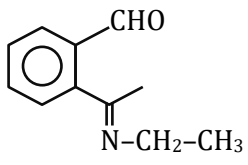
(2)



(3)

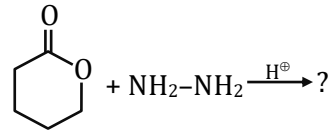


(4)

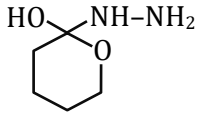


CAA215

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23. 

(1) $HO-(CH_2)_4-\underset{\underset{O}{\|}}{C}-NH-NH_2$

(2) 

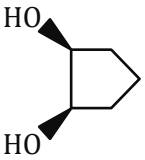
(3) $HO-(CH_2)_3-CH=N-NH_2$

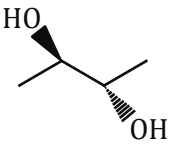
(4) $H_2N-NH-(CH_2)_4-COOH$

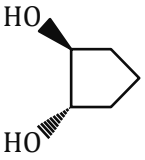
CAA216

24. Which does not react with Acetone

(1) $\begin{matrix}CH_2-OH\\|\\CH_2-OH\end{matrix}$

(2) 

(3) 

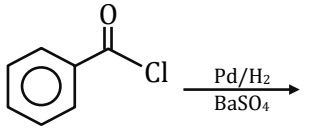
(4) 

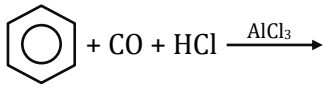
CAA217

25. Which reaction does not gives Aldehyde ?

(1) $CH_3-C\equiv CH + H_2O \xrightarrow[H^+]{Hg^{+2}}$

(2) $Ph-CH_3 + 2Cl_2 \xrightarrow{h\nu} \xrightarrow{H_2O}$

(3) 

(4) 

CAA218

26. Which is maximum soluble in water ?

(1) CH_3-CHO (2) CH_3-COCH_3

(3) CH_3-CH_2-CHO (4) $CH_3-COCH_2-CH_3$

CAA219

27. Which is not correct statement ?

(1) In acidic medium electrophilicity of carbon of carbonyl group is increased.

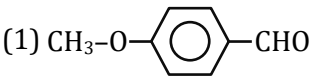
(2) CH_3-CH=NH is more reactive than CH_3-CH=O towards nucleophile

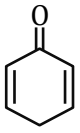
(3) The reaction with $NaHSO_3$ and most of the Ketone is back wards mainly.

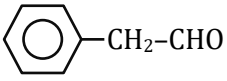
(4) Addition of H–CN on aldehyde/Ketone is catalysed by base to generate strong nucleophile.

CAA220

28. Which compound does not give aldol condensation reaction ?

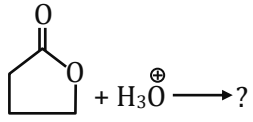
(1) 

(2) 

(3) 

(4) $\begin{matrix}CH_3-CH-CH_2-CHO\\|\\O\end{matrix}$

CAA221

29. 

(1) $HO-(CH_2)_3-COOH$

(2) $HO-(CH_2)_4-OH$

(3) $CH_3-(CH_2)_3-COOH$

(4) $HOO-C(CH_2)_3-COOH$

CAA222

30. CH_3-CH_2-Cl + Mg $\xrightarrow[ether]{dry}$ A $\xrightarrow[(ii)\,H^{\oplus}]{(i)\,CO_2}$ B ?

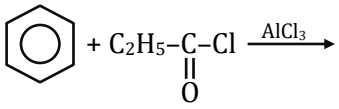
(1) $CH_3-CH_2-\underset{\underset{CH_3}{\|}}{C}-CH_3$ (2) CH_3-CH_2-CHO

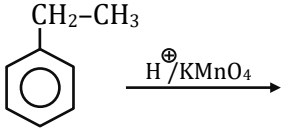
(3) CH_3-CH_2-COOH (4) $CH_3-CH_2-CH_2-OH$

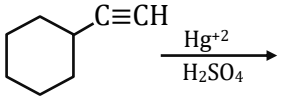
CAA223

31. Which reaction does not give Ketone as major product ?

(1) $2CH_3-\underset{\underset{O}{\|}}{C}-Cl + (C_6H_5-CH_2)_2Cd \longrightarrow$

(2) 

(3) 

(4) 

CAA224

32. Which have maximum dipole moment ?

(1) $H-\underset{\underset{O}{\|}}{C}-H$ (2) CH_3-CHO

(3) $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$ (4) $Ph-\underset{\underset{O}{\|}}{C}-CH_3$

CAA225

33. Oxidation of pentan-2-one by strong oxidising agent gives

(1) Ethanoic acid only

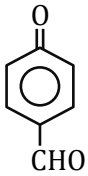
(2) Methanoic acid & butanoic

(3) Ethanoic acid and propanoic acid

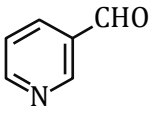
(4) Propanoic acid only

CAA226

34. Which does not give Fehling solution test ?

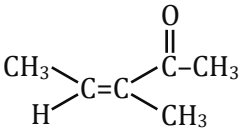
(1) 

(2) $OHC-\left(CHOH\right)_3-\underset{\underset{O}{\|}}{C}-CH_2OH$

(3) 

(4) $CH_3-O-C_6H_4-CHO$

CAA227

35.  A + yellow PPT A

is ?

(1) $CH_3-CHO + O=C(CH_3)-\underset{\underset{O}{\|}}{C}-CH_3$

(2) $CH_3-CH(OH)-C(OH)(CH_3)-COCH_3$

(3) $CH_3-CH=C(CH_3)-COONa$

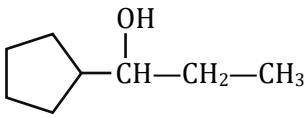
(4) $CH_3-CH=C(COONa)-CH_3$

CAA228

36. Which of the following alcohols can give iodoform reaction?

(1) $CH_3-CH_2-CH(OH)-CH_2-CH_3$

(2) $CH_3-CH(OH)-CH_2-CH_3$

(3) 

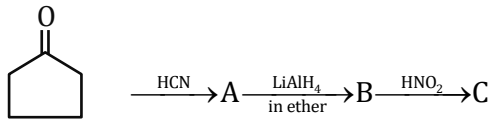
(4) None of these

CAA229

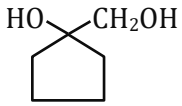
ANSWER KEY

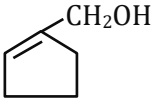
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**3**	**4**	**3**	**3**	**4**	**1**	**3**	**4**	**4**	**1**	**2**	**2**	**3**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**1**	**4**	**1**	**1**	**1**	**3**	**1**	**4**	**1**	**1**	**2**	**1**	**1**	**3**
Question	31	32	33	34	35	36									
Answer	**3**	**4**	**3**	**3**	**4**	**2**									

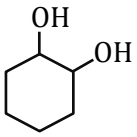
Amines

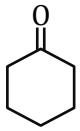
1. 

In above reaction major product (C) is :

(1) 

(2) 

(3) 

(4) 

CAM141

2. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are :

(1) One moles of NaOH and one mole of Br_2

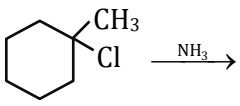
(2) Four mole of NaOH and one mole of Br_2

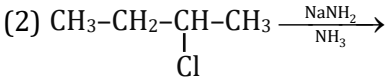
(3) Four moles of NaOH and two moles of Br_2

(4) Two moles of NaOH and two moles of Br_2

CAM142

3. Amine is obtained as major product in

(1) 

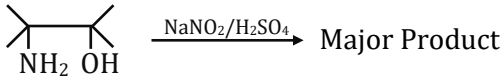
(2) 

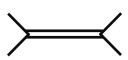
(3) 

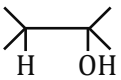
(4) All of these

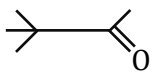
CAM143

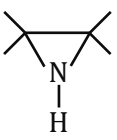
4. The major product of the reaction is



(1) 

(2) 

(3) 

(4) 

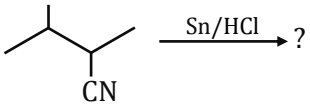
CAM144

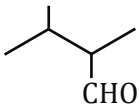
5. The presence of primary amine can be confirmed by these reactions with except :-

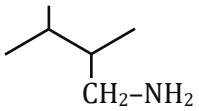
(1) HNO_2 (2) $CHCl_3$+KOH

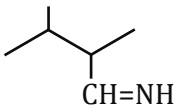
(3) CS_2 and $HgCl_2$ (4) H_2SO_4

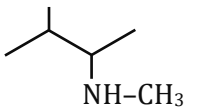
CAM145

6. 

(1) 

(2) 

(3) 

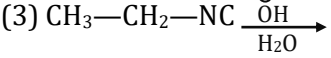
(4) 

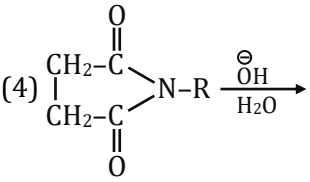
CAM146

7. Which reaction does not give primary amine

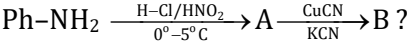
(1) $CH_3—CH_2—Cl + NH_3 \longrightarrow$ excess

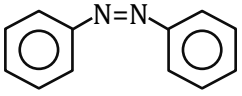
(2) $CH_3—CH_2—Cl + NH_3 \longrightarrow$ excess

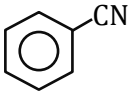
(3) 

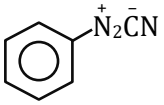
(4) 

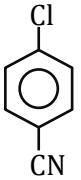
CAM147

8. 

(1) 

(2) 

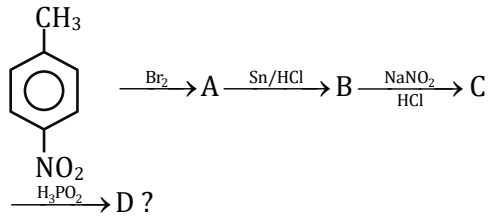
(3) 

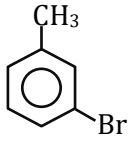
(4) 

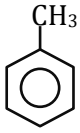
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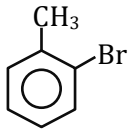
AIM : FREE EDUCATION TO ALL APUL

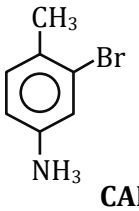
AIM : FREE EDUCATION TO ALL APUL

9. 

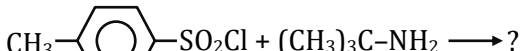
(1) 

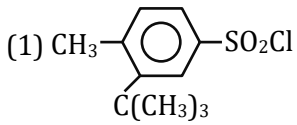
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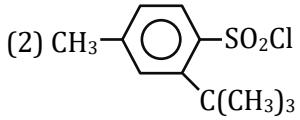
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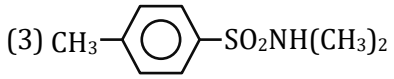
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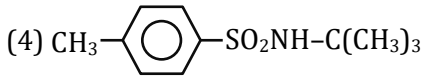
CAM149

10. 

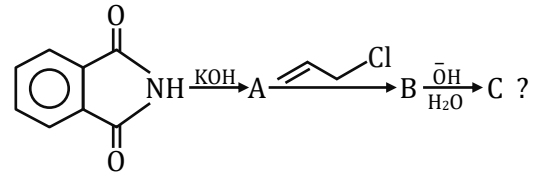
(1) 

(2) 

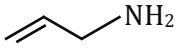
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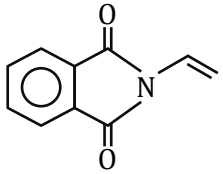
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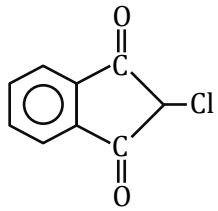
CAM150

11. 

(1) No reaction of A to B

(2) 

(3) 

(4) 

CAM151

12. $C_6H_5-NH_2 + CHCl_3 + \text{Alc. KOH} ?\ A \xrightarrow[H^{\oplus}]{LiAlH_4} B ?$

(1) $C_6H_5-NH_2$ (2) $C_6H_5-NH-CH_3$

(3) $C_6H_5-CH_3$ (4) $C_6H_5-CH_2-NH_2$

CAM152

13. Incorrect statement related to reaction of 1° amine and Hinsberg reagent is

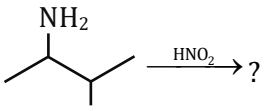
(1) It gives an acid which is soluble in NaOH

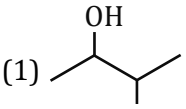
(2) It does not give an acid

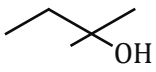
(3) It is nucleophilic substitution reaction

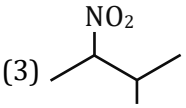
(4) This reaction can be used to distinguish 1° amine and 2° amine.

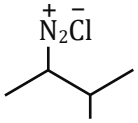
CAM153

14. 

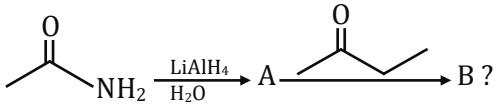
(1) 

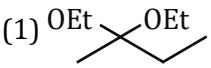
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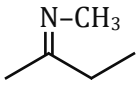
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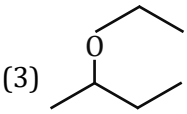
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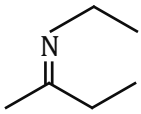
CAM154

15. 

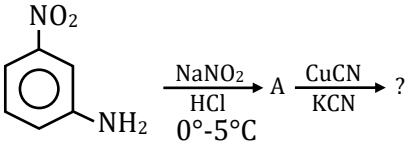
(1) 

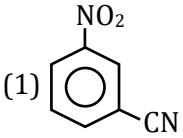
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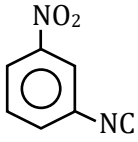
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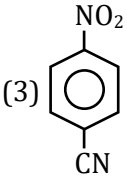
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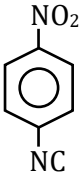
CAM155

16. 

(1) 

(2) 

(3) 

(4) 

CAM156

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**2**	**3**	**3**	**4**	**2**	**1**	**2**	**3**	**4**	**2**	**2**	**2**	**2**	**4**
Question	16														
Answer	**1**														

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